Lecture Notes
Theory of the Finite Element Method

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1. Introduction

These notes are for students from geoscience who are interested in a theoretical background of the Finite Element Methode (FEM) for geodynamic modeling. Due to the wide range of related topics which are involved in this type of modeling (tectonics, rheology, linear algebra, function analysis etc.) some basic knowledge and understanding in these fields is assumed.

The FEM is not a straight forward method to explain. There is unfortunately the need of lots of mathematics. However, with these notes I do not intend to cover the method completely nor do I try to be precise in the sense of a strict and pure mathematical approach. But these shortcomings have to be accepted in order to make the principles of the method as transparent as possible.

For a more advanced introduction of the FEM lots of books are recommended in the reference list in the appendix. These notes only provide you with the basics of the FEM. They do not give a complete overview and are therefore not an excuse for a cross reading in professional textbooks, as for instance in the "FEM-bible" by Zienckiewic and Tayler (1994a and 1994b) where you will find more than 1400 pages in two volumes! This will give you an idea of the complexity and strength of the method. More good textbooks with different approaches to the FEM are Altenbach and Sacharov (1982), Kämmel et al. (1990) and Schwarz (1991). In these lecture notes the aim is to draw a picture of the main steps creating a geodynamic model with FEM. Easy understandable two-dimensional problems using the partial differential equation for the equilibrium of forces (Poisson-equation) with linear elastic rheology will support and accompany the theory.

In most cases the geoscientist will use a commercial FEM-code, the so-called solver (e.g. MSC MARC, ABAQUS, ANSYS, NASTRAN), which is, unfortunately, a black box. Alternatively you can use an open public code, which is to a certain extent written by another geoscientist (e.g. Bird and Kong, 1994). These are often poorly documented and limited to a specific type of problems. A further development is always a very time-consuming task. The alternative is using a commercial FE-software package, which is often provided with a graphical user interface, containing the a pre- and a post-processor (e.g. MSC MENTAT, PATRAN, ABAQUS CAE, Hypermesh von Altair) and the solver. The pre-processor is needed in order to construct the geometry of the model and its discretization, the post-processor is needed for the visualization of the model results. The core of the package is always the solver, which contains the mathematical algorithm for the numerical solution procedures (numerical integration, calculation of the inverse matrix etc.). The reliability of such a software package is high (also the original prices), but the user is always tempted to use the software in a non appropriate way. Solvability or mathematical convergence of the numerical problem does not imply that the result represents the observed process, i.e. the process which has to be modeled. The quality and plausibility strongly depends upon the assumptions and the model input provided and defined by the user. Therefore the results have to be controlled and critically assessed by the user.

Simulation of an observed physical process is a sequence of consecutive steps. The

first one is always the mapping of the physical process on a model process. The next ones are: finding a solution and a time-spatial description which often implies numerical methods. An analytical solution is in the most cases not available. The basic idea of a numerical approximation of the (not available) analytical solution is the transformation of the continuous model description into a discrete one. The equation system will then be solved by a computer with the chosen algorithm (FEM, Finite Differences, Boundary Element Method ....) Therefore, the simulation, i.e. the modeling procedure can be divided into three main steps:

- Definition of the model process and its geometry
- Discretization of the continuum
- Application of a numerical algorithm

Therefore the next chapter after this introduction describes the basic differential equation we have in geoscience and its classification into several groups. Also, some formulas and statements of rheology will be given. The third chapter covers the mathematical treatment of the differential equations before the numerical algorithm of FEM can be applied. The following chapter four gives the introduction into the FEM itself. In the appendix you will find some useful explanations, some mathematical rules which are used in the text and an English-German translation of the main "mathematical" vocabularies for your convenience.

But before starting with the mathematics, I would like to encourage a discussion on the term "model" before we enter into the world of maths. I strongly recommend to start any discussion about a "model" and its results with a short definition and description of the main features of the model type (simplifications, assumptions, type of boundary conditions, used algorithm and also the visualization procedure) before discussing the (geo-)physical meaning, value or result of the model. This will prevent lots of misunderstandings. Setting up a common language is always essential and very helpful. There is a wide range and definition of the term "model" or extended terms like "kinematic model" or "physical model" in all branches of Geoscience. The following is an incomplete list of different model types:

- **Tectonic model**
  Looking at faults (active and inactive ones) and taking into consideration geophysical and geological observations and measurements (gravity, heat flow, seismicity, geological maps, fluvial patterns etc.) a tectonic model can be developed. It does not necessarily give a value as a result (compare with the kinematic model which comes somewhat close to this approach). It is more of an integrative approach which attempts to bring into accordance with a wide range of observed phenomena. (e.g. Sperner et al., 2001)

- **Kinematic model**
  A model which only looks for translation and rotation of rigid plates as e.g. NUVEL-1A from deMets et al. (1994). No internal deformation of the lithospheric plate is assumed, i.e. a rigid body (Euklid) is applied.

- **Dynamic model**
  A difficult term and I have, so far, found three different possibilities of an explanation from literature: The first uses the term dynamic in models where forces are boundary conditions (e.g. slab pull, ridge push). The second definition is according to the time-dependence of the model. A pure elastic or elastic-plastic model is completely time independent and therefore not dynamic, but a model with viscous rheology is time-dependent and can therefore be called dynamic. A third strict definition says, that only models which imply acceleration can be called dynamic. In summary: be careful if somebody talks about a "dynamic model". I dare not decide which definition is the correct one.

- **Mathematical model**
  Often used in geodesy where a vast amount of observation data of the field variable with a high accuracy is available. The mathematical model describes the field by reducing the error towards the observed values (e.g. least square method). This "mathematical model" does not necessarily imply any physics. It aims only at the "perfect" mathematical description of the observed field variable data (see figure 1).In general the model parameter in a geodetic (mathematical) model is estimated through the observed values (field

  variable) by inversion (e.g. least square method, where the model parameter are the unknown coefficients of the mathematical model). In geophysics and geology the model parameters are chosen in order to get the field variable as a result of the model (forward modeling). Of course inverse models are also applied in geology and in geophysics and vice versa forward model are applied in geodesy.

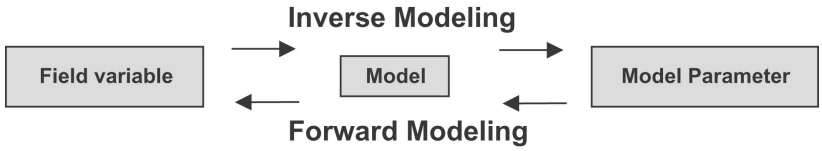


Figure 1: Sketch for the forward and inverse modeling procedure.

- **Forward model**
  This term is used for all modeling approaches where the physical properties and the boundary conditions of the model are given. The field variable, e.g. the displacement field, is calculated from the model. In figure (1) this would be the modeling procedure from the right side to the left. The result of the model is then compared with observed values.

- **Inverse model**
  In contrast to the forward model in the inverse model the field variable has been observed (e.g. gravity or displacement by GPS) and the model parameters are varied in order to find the best fit for the observed field variable. According to figure (1) this would be the procedure from the left to the right side. The

parameters of this mathematical model are varied as long as the average error is minimized.

- **Numerical model**
  Any model which uses a numerical (discrete) method due to the lack of availability of an analytical (continuous) solution. Methods are Finite Differences, Finite Volumes, Finite Elements, Boundary Element etc. .

- **Physical model or analogue model**
  This is a model which you can actually touch. For instance a sandbox model consisting of sirup, plasticine and sand. But this term can also refer to your simplification (your model) of the observed phenomena in nature. E.g. back arc spreading behind subduction zones is observed and the physical model which could give an explanation is the roll-back of the subduction zone. This physical model does not need to consist of formulas, but is only a picture in your mind. This has to be done if you want to convert your physical model into a mathematical formulation in order to solve it e.g. with an numerical algorithm.

- **Static model**
  This term is often used in the commercial FEM software packages. This definition is based on the type of solving algorithm used in the model. If the problem is pure elastic it is called "static model", because the equation of the static equilibrium of forces (accelerations are neglected) has to be solved.

This list is definitely not complete and you will probably find lots of other deviating description in other text books, but it might help you to keep the problem in mind whenever you find yourself in a discussion about your model with another geoscientist.

2. Stating the problem

2.1. Constitutive equations

The following constitutive equations are taken from the textbook "Rheology of the Earth" by Ranalli (1995). Rheology is Greek[1] and can be translated with "science of flow". Hence rheology describes the deformation (flow) of matter under the presence of forces. The two most prominent linear rheological laws, also called constitutive equations, are Hooke's law for elastic deformation and Newton's law for viscous deformation. They relate the stress tensor σ_{ij} to the deformation tensor ϵ_{ij}. A combination of these two can describe many of the deformation processes observed in the lithosphere.

The stress tensor σ_{ij} describes the stress state of any point in a continuous body. The strain tensor ϵ_{kl} describes the deformation, i.e. the change of displacement with space within a continuous body. The relation between the applied forces (or the applied stresses, since stress is measured in force per area with the measure unit Pascal) is described by rheological laws.

2.1.1. Linear elastic rheology

The theoretical model of perfect elasticity postulates that the components of strain at any point within an elastic medium are homogeneous linear functions of the components of stress. The most general description is therefore

$$\sigma_{ij} = C_{ijkl}\epsilon_{kl} \qquad . \tag{1}$$

where C_{ijkl} represent the 81 elastic parameter. Assuming isotropy, i.e. the physical properties do not depend on the direction, this number can be reduced to the two Lame constants λ and μ. With δ the Kronecker symbol and the deviatoric part ϵ'_{ij} of the strain tensor, Hooke law for an isotropic material can be stated as

$$\sigma_{ij} = \lambda\epsilon_{kk}\delta_{ij} + 2\mu\epsilon'_{ij} \qquad . \tag{2}$$

Hooke's law is time-independent, i.e. the reaction of the pure elastic material is instantaneous. As long as the applied forces do not change, the deformation will not change either. The stress tensor can be divided into two parts: The diagonal components of the stress tensor (σ_{11}, σ_{22} and σ_{33}) which describes the lithostatic pressure (see figure 2) and the shear components (off-diagonal components). Since only this deviatoric part (denoted by σ') of the stress tensor produces deformation[2] it is convenient to extract this part and we obtain

$$\sigma'_{ij} = 2\mu\epsilon'_{ij} \tag{3}$$

[1] "$\Pi\alpha\nu\tau\alpha\ \rho\epsilon\iota$ - *everything flows*". This statement is from Herakleitos of Ephesos around 500 v. Chr., who was one of the first natural philosophers trying to explain the processes in nature)

[2] This statement is somewhat tricky and implies that a pure volume change of a material is not called deformation. When $\sigma_{11} = \sigma_{22} = \sigma_{33}$ is applied to rock material, the volume will decrease assuming the rock is compressible. But this "deformation" is not of interest in geodynamic processes. Therefore the term "deformation" is mostly used in the context of deformation due to deviatoric stresses which change the shape of the rock

Instead of the Lame constants the Young modulus E and the Poisson number ν is often used:

$$\begin{aligned} E &= \frac{\mu(3\lambda + 2\mu)}{\lambda + \mu} \\ \nu &= \frac{\lambda}{2(\lambda + \mu)} \\ \mu &= \frac{E}{2(1 + \nu)} \end{aligned} \tag{4}$$

where μ is called the shear modulus often given as G. The range of E and G for Granite for example is in the order of 10-100 GPa (depends on temperature, pressure, texture etc.) whereas ν is around 0.3 (maximum is 0.5 for an incompressible material as, for instance, water).

2.1.2. Linear viscous rheology

A perfect linear viscous material (linar Newton fluid) can be stated as

$$\sigma_{ij} \;=\; -p\delta_{ij} + \lambda' \, \dot{\theta}\delta_{ij} + 2\eta\dot{\epsilon}_{ij} \qquad . \tag{5}$$

where p is the pressure, $\dot{\theta}$ the change in volume, λ' a material constant and η is the viscosity. Since a viscous deformation process is time-dependent the Newton law states the relationship between the applied forces and the deformation rate. When the material is assumed to be incompressible ($\dot{\theta} = 0$) the linear Newton law becomes

$$\sigma_{ij} \;=\; -p\delta_{ij} + 2\eta\dot{\epsilon}_{ij}$$

For the shear component σ'_{ij} of the stress tensor σ_{ij} we therefore obtain

$$\sigma'_{ij} = 2\eta\dot{\epsilon'}_{ij} \qquad . \tag{6}$$

The dot over the strain and shear strain tensor is the time derivative which gives us the strain rate tensor and the shear strain rate tensor respectively. The strain tensor is defined as

$$\epsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) \tag{7}$$

and for the strain rate tensor we receive

$$\begin{aligned} \dot{\epsilon}_{ij} &= \frac{\partial}{\partial t}\left[\frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right] \\ &= \frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right) \end{aligned} \tag{8}$$

where v is the velocity.

2.2. Equation of motion

The equations of equilibrium are applied for a description of an continuous body to be in equilibrium, i.e. the equilibrium of the forces which are acting on and in the body. For an equilibrium the resultant moment about any axis must vanish. This condition implies that the sum of forces in any coordinate direction must vanish. For the description of the forces acting on the surface of a body the Cauchy stress tensor σ_{ij} is used. According to figure (2) this stress tensor is defined as

$$\sigma_{ij} = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{pmatrix} \tag{9}$$

where $\sigma_{11}, \sigma_{22}, \sigma_{33}$ are the normal stresses and $\sigma_{12}, \sigma_{13}, \sigma_{23}, \sigma_{21}, \sigma_{31}, \sigma_{32}$ are the deviatoric stresses[3]. With reference to figure (2) the acting force on any surface of the parallelepiped

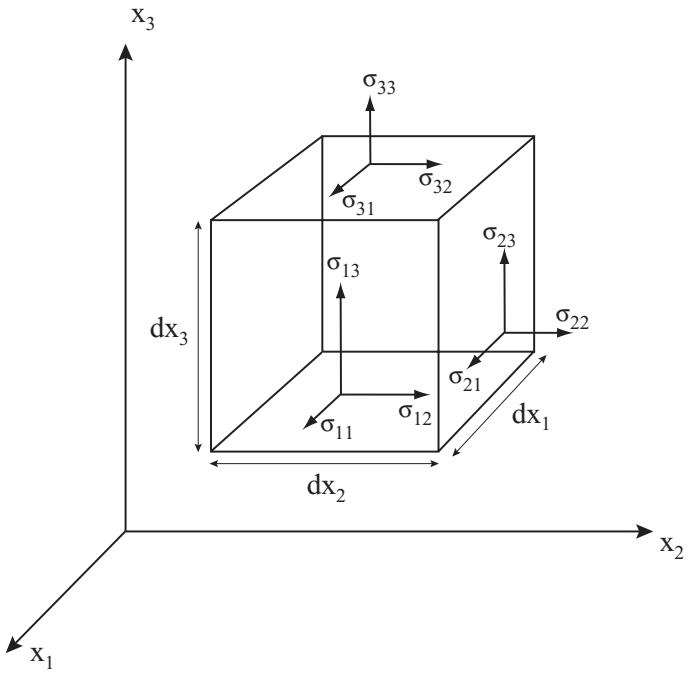


Figure 2: The figure shows the three stress components of the stress tensor σ_{ij} on the three faces of an infinitesimal parallelepiped. The volume of the infinitesimal parallelepiped is defined by its side length dx_1, dx_2 and dx_3 in a cartesian coordinate system

is given by the relevant stress component multiplied by the area of the face. Since there are two parallel faces in each direction of the coordinate system and assuming that the stress components meet certain continuity conditions (first and second derivative

[3]Following figure (2) the first subscript i of the stress tensor denotes the coordinate axis to which the surface is normal and the second subscript j the direction to which the stress is applied to.

towards space must be continuous) stresses (force per area) acting in x_1-direction are

$$\begin{aligned} - \quad & \sigma_{11} dx_2 dx_3 + \left(\sigma_{11} + \frac{\partial \sigma_{11}}{\partial x_1} dx_1 \right) dx_2 dx_3 \\ - \quad & \sigma_{21} dx_1 dx_3 + \left(\sigma_{21} + \frac{\partial \sigma_{21}}{\partial x_2} dx_2 \right) dx_1 dx_3 \\ - \quad & \sigma_{31} dx_1 dx_2 + \left(\sigma_{31} + \frac{\partial \sigma_{31}}{\partial x_3} dx_3 \right) dx_1 dx_2 \quad . \end{aligned} \tag{10}$$

Rearranging this equation gives

$$\left(\frac{\partial \sigma_{11}}{\partial x_1} + \frac{\partial \sigma_{21}}{\partial x_2} + \frac{\partial \sigma_{31}}{\partial x_3} \right) dx_1 dx_2 dx_3 \tag{11}$$

These surface forces have to be in equilibrium with the body forces. With ρ the density of the material and X_1 the body force per unit mass the body forces in x_1-direction is $\rho X_1 dx_1 dx_2 dx_3$. Therefore the equation of equilibrium in x_1-direction is

$$\frac{\partial \sigma_{11}}{\partial x_1} + \frac{\partial \sigma_{21}}{\partial x_2} + \frac{\partial \sigma_{31}}{\partial x_3} + \rho X_1 = 0 \tag{12}$$

The same procedure for the x_2 and x_3-direction gives

$$\frac{\partial \sigma_{12}}{\partial x_1} + \frac{\partial \sigma_{22}}{\partial x_2} + \frac{\partial \sigma_{32}}{\partial x_3} + \rho X_2 = 0 \tag{13}$$

$$\frac{\partial \sigma_{13}}{\partial x_1} + \frac{\partial \sigma_{23}}{\partial x_2} + \frac{\partial \sigma_{33}}{\partial x_3} + \rho X_3 = 0 \tag{14}$$

which can be summarized with index notation as

$$\frac{\partial \sigma_{ij}}{\partial x_i} + \rho X_j = 0 \tag{15}$$

where $i, j = 1, 2, 3$. The stress tensor must be symmetric, i.e. $\sigma_{ij} = \sigma_{ji}$ since the shear stresses have to cancel each other in equation (10). Otherwise a net torque would act on the body and the equilibrium condition would not be fulfilled. The equations of equilibrium state a steady state field problem, since the field variable (for instance the displacement field) is time-independent.

If acceleration a_j can not be neglected,

$$\frac{\partial \sigma_{ij}}{\partial x_i} + \rho X_j = \rho a_j \tag{16}$$

expression (16) is called the equation of motion. Since the field variable (e.g. displacement u) is dependent on time (the acceleration is the second derivative towards time) the equation of motion is a non-steady state field problem, whereas the equations of equilibrium are time-independent and therefore a steady-state field (stationary) problem.

2.2.1. The Navier-Stokes equations

When the increments of the velocity v_j are small the acceleration can also be written in the following form:

$$a_j = \dot{v_j} = \ddot{u_j} = \frac{Dv_j}{Dt} = \frac{D^2u_j}{Dt^2} = \frac{\partial^2 u_j}{\partial t^2} \tag{17}$$

where D is the total derivative, v_j the velocity and t the time. By inserting the definition of the compressible Newton fluid from expression (5) into expression (16) and assuming small velocity increments we receive

$$\frac{\partial}{\partial x_i}\ (-p\delta_{ij} + \lambda'\ \dot{\theta}\delta_{ij} + 2\eta\dot{\epsilon}_{ij}) + \rho X_j = \rho\frac{\partial^2 u_j}{\partial t^2} \qquad . \tag{18}$$

These equations of motion are called the Navier-Stokes equations (for a compressible fluid with linear Newton viscosity). Neglecting the acceleration, which is reasonable for most of the tectonic processes, and assuming incompressibility ($\dot{\theta} = 0$) we receive

$$\frac{\partial}{\partial x_i}(-p\delta_{ij} + 2\eta\dot{\epsilon}_{ij}) + \rho X_j = 0 \qquad . \tag{19}$$

Multiplying expression (19) with ρ^{-1} and inserting expression (8) for $\dot{\epsilon}_{ij}$ leads to

$$\begin{aligned} & -\frac{1}{\rho}\frac{\partial p}{\partial x_i} + \frac{1}{\rho}\eta\frac{\partial}{\partial x_i}\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right) + X_j = 0 \\ \Rightarrow\quad & -\frac{1}{\rho}\frac{\partial p}{\partial x_i} + \frac{1}{\rho}\eta\ \bigtriangledown^2 u_j + X_j = 0 \end{aligned} \tag{20}$$

2.2.2. Equation of equilibrium

A body in equilibrium implies that accelerations can be neglected in expression (16). By inserting the Hook law for the shear component from expression (3) and expression (7) for the strain tensor we receive

$$\begin{aligned} & \frac{\partial}{\partial x_i}(2\mu\epsilon'_{ij}) + \rho X_j = 0 \\ \Rightarrow\quad & \mu\frac{\partial {u_j}^2}{\partial x_i^2} + \rho X_j = 0 \\ \Rightarrow\quad & \mu\ \bigtriangledown^2 u_j + \rho X_j = 0 \\ \Rightarrow\quad & \mu\ \triangle\ u_j + \rho X_j = 0 \end{aligned} \tag{21}$$

assuming that μ, the shear modulus (also denoted by G in the literature) is homogeneous. This is a Poisson equation which is a linear partial differential equation of 2nd order.

2.3. Mathematical classification of linear partial differential equations of 2nd order

Most of the differential equations in geoscience are linear partial differential equations (PDE's) of 2nd order. In two dimensions they can be stated in a general form as

$$A\frac{\partial^2 u}{\partial x_1^2} + 2B\frac{\partial^2 u}{\partial x_1 \partial x_2} + C\frac{\partial^2 u}{\partial x_2^2} = F \tag{22}$$

where $A, B, C,\ F$ and u are functions of the coordinates x_1, x_2. The mathematical classification differentiates between parabolic, hyperbolic and elliptical PDE's[4] according to the value of $D = AC - B^2$.

$$\begin{aligned} parabolic &\quad \Leftrightarrow \quad D = 0 \\ hyperbolic &\quad \Leftrightarrow \quad D < 0 \\ elliptic &\quad \Leftrightarrow \quad D > 0 \end{aligned} \tag{23}$$

According to their physical meaning the PDE's are classified into two groups: the time-independent steady state field problems and time-dependent non-steady state field problems.

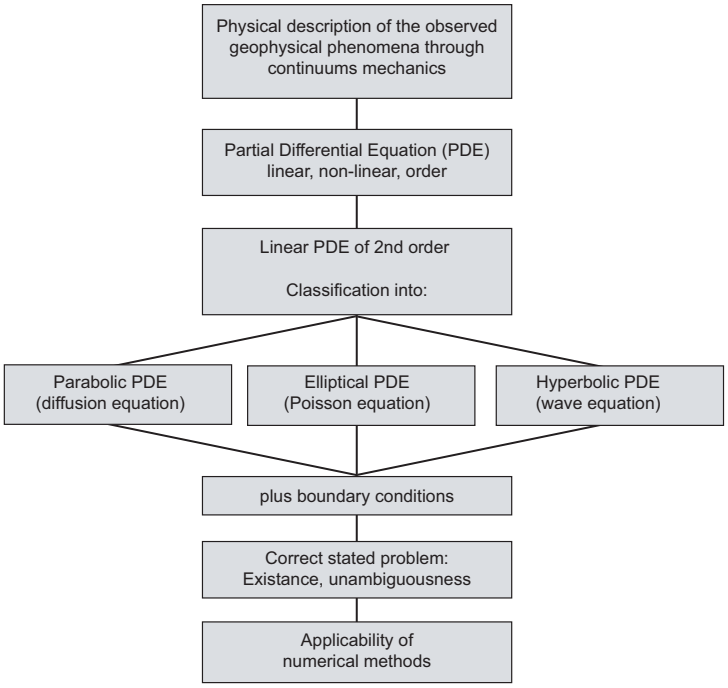


Figure 3: Flowchart of the classification of linear Partial Differential Equations (PDE) of 2nd order and their general solving scheme.

[4]This comes from the planar analytical geometry where $ax^2 + 2bxy + cy^2 + dx + ey + f = 0$ describes a parabola, hyperbola or ellipse in a case if $d = ac - b^2 = 0, < 0$, or > 0

2.4. Physical classification of linear partial differential equations of 2nd order

The most general description of linear partial differential equations of 2nd order in a two-dimensional problem with anisotropic material behavior is

$$\frac{\partial}{\partial x_1}\left(a_1(x_i,t)\frac{\partial u}{\partial x_1}\right)+\frac{\partial}{\partial x_2}\left(a_2(x_i,t)\frac{\partial u}{\partial x_2}\right)= f(x_i,t)+b(x_i,t)\frac{\partial u}{\partial t}+c(x_i,t)\frac{\partial^2 u}{\partial t^2} \qquad (24)$$

where u is either a scalar (e.g. temperature) or a vector (e.g. displacement) and the functions $a_1(x_1,x_2)$ and $a_2(x_1,x_2)$ describe the material properties ($i,j=1,2$ according to the spatial dimensions).

According to figure (4) the PDE is defined in the area S with the border C. By

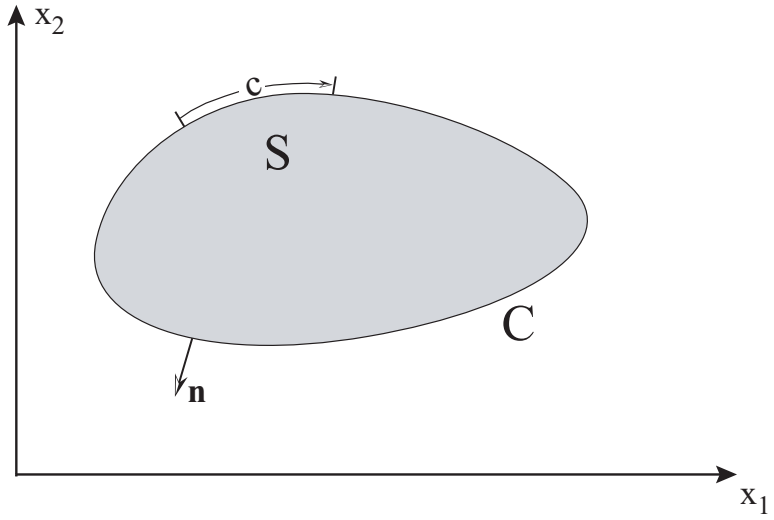


Figure 4: Sketch of the surface S where the PDE is described. On the border C, which is divided into two parts C_1 and C_2, the boundary conditions are prescribed. $\mathbf{n}$ is the outer normal of the border and c is the circular length.

dividing the border C into two segments C_1 and C_2, various boundary conditions can be described at the border. The function $u(x_1,x_2)$ has to fulfill certain values which are prescribed by the functions $\beta(c),\chi(c)$ and $\gamma(c)$ of the circular length c.

$$u(c) = \chi(c) \quad on\ C_1 \quad \text{Dirichlet boundary condition} \qquad (25)$$

$$\frac{\partial u(c)}{\partial \mathbf{n}(c)}+\beta(c)\cdot u(c) = \gamma(c) \quad on\ C_2 \quad \text{Cauchy boundary condition} \qquad (26)$$

where $\mathbf{n}(c)$ is the outer normal of the border C. The Cauchy boundary condition can be reduced to the following Neumann boundary condition

$$\frac{\partial u(c)}{\partial \mathbf{n}(c)} = 0 \quad on\ C_2 \quad \text{Neumann boundary condition} \qquad (27)$$

in case of $\beta(c) = \gamma(c) = 0$. Since $\mathbf{n}(c)$ is the outer normal of the border C, the Neumann boundary condition denotes that $u(c)$ is not allowed to change perpendicular to the border C of the surface S.

Also, the boundary conditions are now time-dependent. Additionally an initial value, which describes the state of the field variable at time t_0, has to be given. Three varieties of expression (24) for a general partial differential equation (PDE) can be distinguished and are well known in geoscience.

Type	**Mathematical classification**	**Physical classification Example**
$c(x_i, t) = 0$	parabolic PDE	non-steady state diffusion equation
$b(x_i, t) = f(x_i, t) = 0$	hyperbolic PDE	non-steady state Helmholtz or wave equation in elasto-dynamics
$b(x_i, t) = c(x_i, t)$=0	elliptical PDE	steady state equation of equlibrium potential equation

For steady state field problems the PDE only depends on u and their spatial derivatives, where u is either a scalar (e.g. temperature) or a vector (e.g. displacement). The functions $a_1(x_1, x_2)$ and $a_2(x_1, x_2)$ describe the material properties. In an anisotropic non-homogeneous material the general expression for a steady-state field problem in two dimensions is:

$$\frac{\partial}{\partial x_1}\left(a_1(x_1, x_2)\frac{\partial u}{\partial x_1}\right)+\frac{\partial}{\partial x_2}\left(a_2(x_1, x_2)\frac{\partial u}{\partial x_2}\right)= f(x_1, x_2) \tag{28}$$

An geoscientific example is the calculation of the geothermal gradient with depth. The field variable is the $u = T(x_1, x_2)$, the material property is the thermal diffusivity κ and the function $f(x_1, x_2$ is the heat production due to radioactive decay. Assuming that κ is homogeneous expression (28)

$$a_1 = a_2 = \kappa = 10^{-6} \tag{29}$$

the PDE is the Poisson equation

$$10^{-6}\frac{\partial^2 T}{\partial x_1^2} + 10^{-6}\frac{\partial^2 T}{\partial x_2^2} = 10^{-6} \triangle T = f(x_i) \tag{30}$$

In the case of no heat production ($f(x_1, x_2) = 0$) and no dependence in x_1-direction (x_2 beeing the vertical direction of the lithosphere) the Poisson equation is called the Laplace equation

$$\frac{\partial^2 T}{\partial x_2^2} = 0 \tag{31}$$

With boundary conditions at the bottom of the lithosphere ($T_m = 1500$ K; melting temperature of lithospheric rocks) and the Earth surface ($T_0 = 280K$) the Laplace equation can be solved by integration.

3. Approximation of the PDE solutions

In most cases it is not possible to find an analytical solution for the PDE. As a consequence only an approximation of the solution can be found by applying an appropriate numerical algorithm. Each of the examples given in figure **??** needs a discretisation of the model area. There is always the fight between accuracy and flexibility (capability to describe irregular geometries and complicated boundary conditions) of the numerical method. E.g. the accuracy of the Finite Difference method is higher compared to the FEM, but the flexibility of the FEM is much better. As a preparation before we

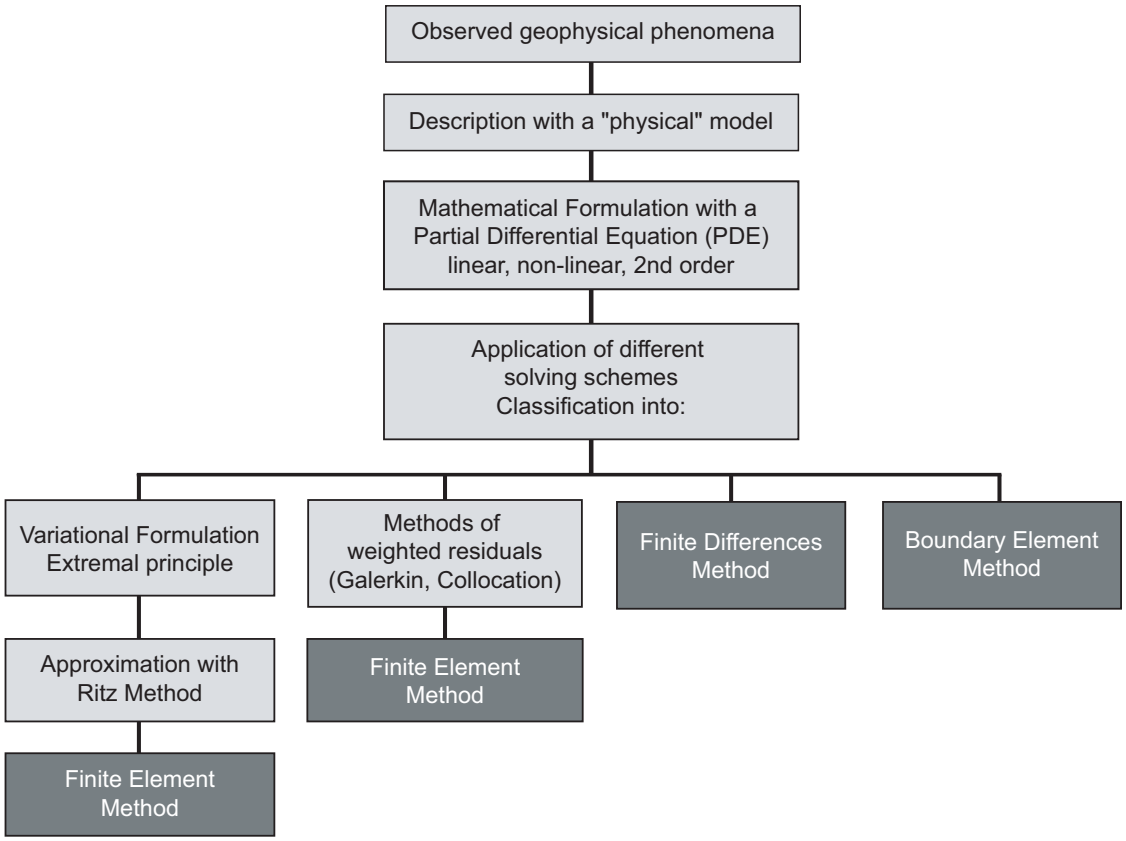


Figure 5: Flowchart of the classification of Partial Differential Equations (PDE) of 2nd order and their general solving scheme

apply the Finite Element Method we have to rewrite the PDE. This can be achieved in two different ways. The variational formulation with subsequent Ritz method or the Galerkin method (method of weighed residuals). Both approaches (Ritz and Galerkin) seek numerical approximations of the wanted solution for the PDE and they both lead to the same set of equations, but still a solution is only found for simple geometries of the model area. However both approaches will bring us directly to the world of Finite Elements.

3.1. The Variational Principle

In contrast to an extremum calculation for functions, i.e. looking for a value of the function where the function has an extremum (minimum or maximum), in variational calculus one seeks a function which minimizes a functional. A functional is an equation which is not dependent on coordinates anymore but on the functions. Functions are now the variables.

Let us consider the functions a_1, a_2, a_3, u, f and g in a given space $V \subset \mathbf{R^3}$. S is the surface of V, S_1 and S_2 are parts of S and $S_1 \cup S_2 = S$. The functions $\beta(s)$ and $\gamma(s)$ describe the boundary conditions and are therefore defined on the surface S of V. The vector $\mathbf{n}$ is the outer normal onto the surface S_2. The function $u(x_i)$ with $i = 1, 2, 3$ which then minimizes the functional

$$\begin{aligned} I(u) &= \int_V \left[\frac{1}{2} \left(a_1 \left(\frac{\partial u}{\partial x_1} \right)^2 + a_2 \left(\frac{\partial u}{\partial x_2} \right)^2 + a_3 \left(\frac{\partial u}{\partial x_3} \right)^2 \right) - \frac{1}{2} f u^2 - g u \right] dV \\ &+ \oint_S \left[\frac{1}{2} \beta(s) \cdot u^2(s) - \gamma(s) \cdot u(s) \right] dS \end{aligned} \tag{32}$$

also solves the differential equation

$$\frac{\partial}{\partial x_1} a_1 \left(\frac{\partial u}{\partial x_1} \right) + \frac{\partial}{\partial x_2} a_2 \left(\frac{\partial u}{\partial x_2} \right) + \frac{\partial}{\partial x_3} a_3 \left(\frac{\partial u}{\partial x_3} \right) + f u + g = 0 \tag{33}$$

with the Dirichlet boundary condition $u(s) = \chi(s)$ on S_1 and the general Cauchy boundary condition on S_2

$$a_1 \frac{\partial u}{\partial x_1} n_1 + a_2 \frac{\partial u}{\partial x_2} n_2 + a_3 \frac{\partial u}{\partial x_3} n_3 + \beta(s) = \gamma(s) \qquad . \tag{34}$$

Every differential equation, where the linear differential operator L is positive definite, can always be stated as a variational formulation[5]. For the special case where $\beta(s) = \gamma(s) = 0$, the general Cauchy boundary condition is called the Neumann boundary condition.

3.2. Proof of the Variational Principle

In order to solve the variational problem the first variation $\delta I(u)$ of the integral expression $I(u)$ (which is the first term of the Taylor expansions for functionals) has to vanish. Or in other words, to find a minimum of the functional $I(u)$ This leads to

$$\begin{aligned} \delta I(u) &= \int_V \left[\frac{1}{2} \left(a_1 \delta \left[\left(\frac{\partial u}{\partial x_1} \right)^2 \right] + a_2 \delta \left[\left(\frac{\partial u}{\partial x_2} \right)^2 \right] + a_3 \delta \left[\left(\frac{\partial u}{\partial x_3} \right)^2 \right] \right) - \frac{1}{2} f \delta(u^2) - g \delta u \right] dV \\ &+ \oint_S \left[\frac{1}{2} \beta \delta(u^2) - \gamma \delta u \right] dS \;=\; 0 \qquad . \end{aligned} \tag{35}$$

[5] If L is self-adjoint and elliptic, L is positive definite (see also appendix A). The differential operator Laplace $\triangle$ and the differential operator Nabla ∇ operator are positive definite

Applying the calculus rules for the first variation (see appendix A) on equation (35) we receive

$$\begin{aligned}\delta I(u) &= \int_V \left[a_1 \frac{\partial u}{\partial x_1}\frac{\partial}{\partial x_1}\delta u + a_2 \frac{\partial u}{\partial x_2}\frac{\partial}{\partial x_2}\delta u + a_3 \frac{\partial u}{\partial x_3}\frac{\partial}{\partial x_3}\delta u - (fu+g)\delta u \right] dV \\ &+ \oint_S \left[(\beta u - \gamma)\delta u \right] dS \;=\; 0 \end{aligned} \tag{36}$$

and applying the integration by parts (see appendix A, expression 171) on every of the first three terms of the volume integral (36) we receive for every term

$$\begin{aligned} &\int_V \left[a_1 \frac{\partial u}{\partial x_1}\frac{\partial}{\partial x_1}\delta u + a_2 \frac{\partial u}{\partial x_2}\frac{\partial}{\partial x_2}\delta u + a_3 \frac{\partial u}{\partial x_3}\frac{\partial}{\partial x_3}\delta u \right] dV \\ = &\int_V \frac{\partial}{\partial x_1}\left(a_1 \frac{\partial u}{\partial x_1}\delta u \right) dV - \int_V \frac{\partial}{\partial x_1}\left(a_1 \frac{\partial u}{\partial x_1} \right)\delta u \; dV \\ + &\int_V \frac{\partial}{\partial x_2}\left(a_2 \frac{\partial u}{\partial x_2}\delta u \right) dV - \int_V \frac{\partial}{\partial x_2}\left(a_2 \frac{\partial u}{\partial x_2} \right)\delta u \; dV \\ + &\int_V \frac{\partial}{\partial x_3}\left(a_3 \frac{\partial u}{\partial x_3}\delta u \right) dV - \int_V \frac{\partial}{\partial x_3}\left(a_3 \frac{\partial u}{\partial x_3} \right)\delta u \; dV \quad . \end{aligned} \tag{37}$$

Applying the integral sentence of Gauss (see appendix A, expression (176)) on the three positive volume integrals on the right side of expression (37) we receive

$$\begin{aligned} &\oint_S a_1 \frac{\partial u}{\partial x_1} n_1 \delta u \; dS - \int_V \frac{\partial}{\partial x_1}\left(a_1 \frac{\partial u}{\partial x_1} \right)\delta u \; dV \\ + &\oint_S a_2 \frac{\partial u}{\partial x_2} n_2 \delta u \; dS - \int_V \frac{\partial}{\partial x_2}\left(a_2 \frac{\partial u}{\partial x_2} \right)\delta u \; dV \\ + &\oint_S a_3 \frac{\partial u}{\partial x_3} n_3 \delta u \; dS - \int_V \frac{\partial}{\partial x_3}\left(a_3 \frac{\partial u}{\partial x_3} \right)\delta u \; dV \quad . \end{aligned} \tag{38}$$

Rearranging and summarizing expression (38) leads directly towards

$$\begin{aligned} &\oint_S \left[a_1 \frac{\partial u}{\partial x_1} n_1 + a_2 \frac{\partial u}{\partial x_2} n_2 + a_3 \frac{\partial u}{\partial x_3} n_3 \right] \delta u \; dS \\ &- \int_V \left[\frac{\partial}{\partial x_1}\left(a_1 \frac{\partial u}{\partial x_1} \right) + \frac{\partial}{\partial x_2}\left(a_2 \frac{\partial u}{\partial x_2} \right) + \frac{\partial}{\partial x_3}\left(a_3 \frac{\partial u}{\partial x_3} \right) \right] \delta u \; dV \quad . \end{aligned} \tag{39}$$

Inserting expression (39) into expression (36) we receive

$$\begin{aligned} &- \int_V \left[\frac{\partial}{\partial x_1} a_1 \left(\frac{\partial u}{\partial x_1} \right) + \frac{\partial}{\partial x_2} a_2 \left(\frac{\partial u}{\partial x_2} \right) + \frac{\partial}{\partial x_3} a_3 \left(\frac{\partial u}{\partial x_3} \right) + fu + g \right] \delta u \; dV \\ &+ \oint_S \left[a_1 \frac{\partial u}{\partial x_1} n_1 + a_2 \frac{\partial u}{\partial x_2} n_2 + a_3 \frac{\partial u}{\partial x_3} n_3 + \beta u - \gamma \right] \delta u \; dS \;=\; 0 \quad . \end{aligned} \tag{40}$$

Due to the condition that these integrals have to vanish for all variations of δu of u, the expression under both integrals must vanish, i.e. they fulfill the stated differential equation (33). If one of the integrals would not equal zero it would be possible to find such δu that the left hand side of expression (40) would not equal zero. The function $u(x_i)$ which minimizes expression (32) and solves the partial differential equation (33) with the boundary conditions stated in (34).

3.3. 2-D example for the Variational Principle

Let us assume that $\beta = \gamma = 0$, $g = 1$, $f = 0$, $a_1 = 1$, $a_2 = 2$ in a two-dimensional problem with the cartesian coordinates x_1 and x_2. Filling this into expression (33) we will then receive:

$$\frac{\partial^2 u}{\partial x_1^2} + 2\,\frac{\partial^2 u}{\partial x_2^2} + 1 \;=\; 0 \tag{41}$$

with the Dirichlet boundary condition $u = \chi$ on C_1 and the general Cauchy boundary condition on C_2

$$\frac{\partial u}{\partial x_1} n_1 + 2\,\frac{\partial u}{\partial x_2} n_2 = 0 \tag{42}$$

According to expression (32) the variational formulation of this PDE leads to the following integral expression

$$I(u) \;=\; \int_S \left(\frac{1}{2} \left[\left(\frac{\partial u}{\partial x_1} \right)^2 + 2 \left(\frac{\partial u}{\partial x_2} \right)^2 \right] + u \right) dS \tag{43}$$

In order to solve the variational formulation the first variation $\delta I(u)$ of the integral expression $I(u)$ (which is the first term of the Taylor expansions for functionals) has to vanish. This leads to

$$\delta I(u) = \frac{\partial I(u)}{\partial u} = \int_S \left[\frac{1}{2} \left(\delta \left[\left(\frac{\partial u}{\partial x_1} \right)^2 \right] + 2\delta \left[\left(\frac{\partial u}{\partial x_2} \right)^2 \right] \right) - \delta u \right] dS = 0 \tag{44}$$

Applying the calculus rules for the first variation (see appendix A, expression (170)) on equation (44) we receive

$$\delta I(u) \;=\; \int_S \left[\frac{\partial u}{\partial x_1} \frac{\partial}{\partial x_1} \delta u + 2 \frac{\partial u}{\partial x_2} \frac{\partial}{\partial x_2} \delta u - \delta u \right] dS = 0 \tag{45}$$

Applying integration by parts (see appendix A, expression(171)) on the first two terms of the surface integral (45) we receive for every term

$$\begin{aligned} & \int_S \left[\frac{\partial u}{\partial x_1} \frac{\partial}{\partial x_1} (\delta u) + 2 \frac{\partial u}{\partial x_2} \frac{\partial}{\partial x_2} (\delta u) \right] dA \\ = \; & \int_S \left[\frac{\partial}{\partial x_1} \left(\frac{\partial u}{\partial x_1} \delta u \right) \right] dS - \int_S \left[\frac{\partial}{\partial x_1} \left(\frac{\partial u}{\partial x_1} \right) \delta u \right] dS \\ + \; & \int_S \left[\frac{\partial}{\partial x_2} \left(2 \frac{\partial u}{\partial x_2} \delta u \right) \right] dS - \int_S \left[\frac{\partial}{\partial x_2} \left(2 \frac{\partial u}{\partial x_2} \right) \delta u \right] dS \end{aligned} \tag{46}$$

Applying the integral sentence of Gauss (see appendix A, expression (176) the two positive surface integrals of (46) can be changed into a line integral. Equation (46) then becomes

$$\begin{aligned} & \int_S \frac{\partial}{\partial x_1}\left(\frac{\partial u}{\partial x_1}\delta u\right)\, dS - \int_S \frac{\partial}{\partial x_1}\left(\frac{\partial u}{\partial x_1}\right)\delta u\, dS \\ + & \int_S \frac{\partial}{\partial x_2}\left(2\frac{\partial u}{\partial x_2}\delta u\right)\, dS - \int_S \frac{\partial}{\partial x_2}\left(2\frac{\partial u}{\partial x_2}\right)\delta u\, dS \\ = & \oint_C \frac{\partial u}{\partial x_1} n_1 \delta u\, dC - \int_S \frac{\partial}{\partial x_1}\left(\frac{\partial u}{\partial x_1}\right)\delta u\, dS \\ + & \oint_C 2\frac{\partial u}{\partial x_2} n_2 \delta u\, dC - \int_S \frac{\partial}{\partial x_2}\left(2\frac{\partial u}{\partial x_2}\right)\delta u\, dS \end{aligned} \tag{47}$$

and by rearranging this leads directly to

$$\begin{aligned} = & \oint_C \frac{\partial u}{\partial x_1} n_1 \delta u\, dC + \oint_C 2\, \frac{\partial u}{\partial x_2} n_2 \delta u\, dC \\ - & \int_S \frac{\partial}{\partial x_1}\left(\frac{\partial u}{\partial x_1}\right)\delta u\, dS - \int_S \frac{\partial}{\partial x_2}\left(2\, \frac{\partial u}{\partial x_2}\right)\delta u\, dS \end{aligned} \tag{48}$$

$$\begin{aligned} = & \oint_C \left[\frac{\partial u}{\partial x_1} n_1 + 2\, \frac{\partial u}{\partial x_2} n_2\right]\delta u\, dC \\ - & \int_S \left[\frac{\partial}{\partial x_1}\left(\frac{\partial u}{\partial x_1}\right) + \frac{\partial}{\partial x_2}\left(2\, \frac{\partial u}{\partial x_2}\right)\right]\delta u\, dS \end{aligned} \tag{49}$$

If expression (49) is substituted in (45) we receive

$$\begin{aligned} & -\int_S \left[\frac{\partial}{\partial x_1}\left(\frac{\partial u}{\partial x_1}\right) + \frac{\partial}{\partial x_2} 2\left(\frac{\partial u}{\partial x_2}\right) + 1\right]\delta u\, dS \\ & \quad + \oint_C \left[\frac{\partial u}{\partial x_1} n_1 + 2\, \frac{\partial u}{\partial x_2} n_2\right]\delta u\, dC \;=\; 0 \end{aligned} \tag{50}$$

Due to the condition that these integrals have to vanish for all variations of δu of u, the expression under the integrals must vanish, i.e. they fulfill the stated differential equation (41). That is, the function $u(x_i)$ which minimizes equation (43) also solves the partial differential equation (41). But also for the variational formulation of the PDE, where the order of the differential equation is reduced by one, it is difficult, or even impossible, to find an analytical solution.

3.4. Ritz Method

As stated in the previous chapter an analytical solution is difficult to receive for the PDE and its variational formulation. Therefore an approximation procedure must be applied. Let us consider a set of linear independent functions

$$\phi_0(x_i), \phi_1(x_i) \ldots, \phi_m(x_i) \qquad . \tag{51}$$

As the problem is not solvable in $u(x_i), i = 1, 2, 3$ an approximation $u_h(x_i)$ as a linear combination of $\phi_k(x_i)$ is chosen.

$$\begin{aligned} u_h(x_i) &= \sum_{k=0}^{m} \alpha_k \ \phi_k(x_i) \\ &= \alpha_0 \phi_0(x_i) + \sum_{k=1}^{m} \alpha_k \ \phi_k(x_i) \qquad . \end{aligned} \tag{52}$$

It is reasonable that $\phi_0(x_i)$ fulfills the boundary conditions and that all other $\phi_j(x_i)$ for $j = 1, ..., m$ vanish at the boundary. Substituting this approximation into equation (32) one gets

$$\begin{aligned} I(\alpha_0, \ldots, \alpha_m) &= \int_V \left[\frac{a_1}{2} \left(\sum_{k=0}^{m} \alpha_k \frac{\partial \phi_k(x_i)}{\partial x_1} \right)^2 + \frac{a_2}{2} \left(\sum_{k=0}^{m} \alpha_k \frac{\partial \phi_k(x_i)}{\partial x_2} \right)^2 + \frac{a_3}{2} \left(\sum_{k=0}^{m} \alpha_k \frac{\partial \phi_k(x_i)}{\partial x_3} \right)^2 \right. \\ &- \left. \frac{f}{2} \left(\sum_{k=0}^{m} \alpha_k \ \phi_k(x_i) \right)^2 - g \left(\sum_{k=0}^{m} \alpha_k \ \phi_k(x_i) \right) \right] \ dV \\ &+ \oint_S \left[\frac{\beta}{2} \left(\sum_{k=0}^{m} \alpha_k \ \phi_k(x_i) \right)^2 - \gamma \left(\sum_{k=0}^{m} \alpha_k \ \phi_k(x_i) \right) \right] \ dS \end{aligned} \tag{53}$$

where $\alpha_0 = 1$ and $\phi_0(x_i)$ fulfils the boundary conditions on S. Therefore only the remaining m functions $\phi_m(x_i)$ have to be taken into consideration. In order to make this expression extremal one has to set the first partial differentiations to zero (with $j = 1, ..., m$ because α_0 is per definition already known)

$$\begin{aligned} \delta I(\alpha_j) = \frac{\partial I}{\partial \alpha_j} &= \int_V \left[a_1 \sum_{k=0}^{m} \alpha_k \frac{\partial \phi_k(x_i)}{\partial x_1} \frac{\partial \phi_j(x_i)}{\partial x_1} \right. \\ &+ \left. a_2 \sum_{k=0}^{m} \alpha_k \frac{\partial \phi_k(x_i)}{\partial x_2} \frac{\partial \phi_j(x_i)}{\partial x_2} + a_3 \sum_{k=0}^{m} \alpha_k \frac{\partial \phi_k(x_i)}{\partial x_3} \frac{\partial \phi_j(x_i)}{\partial x_3} \right] \ dV \\ &- \int_V \left[f \left(\sum_{k=0}^{m} \alpha_k \phi_k(x_i) \phi_j(x_i) \right) + g \ \phi_j(x_i) \right] \ dV \\ &+ \oint_S \left[\beta \left(\sum_{k=0}^{m} \alpha_k \phi_k(x_i) \phi_j(x_i) \right) - \gamma \ \phi_j(x_i) \right] \ dS \ = \ 0 \end{aligned} \tag{54}$$

3.5. 2-D example for the Ritz Method

Again as in the 2-D example for the Variational Method ($\beta = \gamma = 0$, $g = 1$, $f = 0$, $a_1 = 1$, $a_2 = 2$) the given PDE is

$$\frac{\partial^2 u}{\partial x_1^2} + 2 \ \frac{\partial^2 u}{\partial x_2^2} + 1 \ = \ 0 \tag{55}$$

with the Dirichlet boundary condition $u = \chi$ on C_1 and the general Cauchy boundary condition on C_2

$$\frac{\partial u}{\partial x_1} n_1 + 2 \frac{\partial u}{\partial x_2} n_2 = 0 \qquad . \tag{56}$$

From expression (54) we receive with the above stated definitions for β, γ, g, f, a_1 and a_2 for this two dimensional problem

$$\frac{\partial I}{\partial \alpha_j} = \int_S \left[\sum_{k=0}^{m} \alpha_k \frac{\partial \phi_k(x_i)}{\partial x_1} \frac{\partial \phi_j(x_i)}{\partial x_1} + 2 \sum_{k=0}^{m} \alpha_k \frac{\partial \phi_k(x_i)}{\partial x_2} \frac{\partial \phi_j(x_i)}{\partial x_2} - \phi_j(x_i) \right] dS = 0 \tag{57}$$

which can be rewritten into

$$\begin{aligned} & \int_S \left[\sum_{k=1}^{m} \alpha_k \frac{\partial \phi_k(x_i)}{\partial x_1} \frac{\partial \phi_j(x_i)}{\partial x_1} + 2 \sum_{k=1}^{m} \alpha_k \frac{\partial \phi_k(x_i)}{\partial x_2} \frac{\partial \phi_j(x_i)}{\partial x_2} \right] dS \\ + & \int_S \left[\alpha_0 \frac{\partial \phi_0(x_i)}{\partial x_1} \frac{\partial \phi_j(x_i)}{\partial x_1} + 2\, \alpha_0 \frac{\partial \phi_0(x_i)}{\partial x_2} \frac{\partial \phi_j(x_i)}{\partial x_2} - \phi_j(x_i) \right] dS = 0 \end{aligned} \tag{58}$$

Rearranging expression (58) and taking into account that $\alpha_0 = 1$ is defined in order to fulfill the boundary conditions [$\phi_0(x_i)$ describes the boundary conditions at the boundary where all other function $\phi_j(x_i)$ vanish] we receive

$$\begin{aligned} & \int_S \left[\sum_{k=1}^{m} \alpha_k \frac{\partial \phi_k(x_i)}{\partial x_1} \frac{\partial \phi_j(x_i)}{\partial x_1} + 2 \sum_{k=1}^{m} \alpha_k \frac{\partial \phi_k(x_i)}{\partial x_2} \frac{\partial \phi_j(x_i)}{\partial x_2} \right] dS \\ + & \int_S \left[\frac{\partial \phi_0(x_i)}{\partial x_1} \frac{\partial \phi_j(x_i)}{\partial x_1} + 2 \frac{\partial \phi_0(x_i)}{\partial x_2} \frac{\partial \phi_j(x_i)}{\partial x_2} - \phi_j(x_i) \right] dS \;=\; 0 \qquad . \end{aligned} \tag{59}$$

This leads to a set of linear equations of the j unknown parameter α_j

$$a_{jk} := \int_S \left[\frac{\partial \phi_k(x_i)}{\partial x_1} \frac{\partial \phi_j(x_i)}{\partial x_1} + 2 \frac{\partial \phi_k(x_i)}{\partial x_2} \frac{\partial \phi_j(x_i)}{\partial x_2} \right] dS \tag{60}$$

$$b_j := - \int_S \left[\frac{\partial \phi_0(x_i)}{\partial x_1} \frac{\partial \phi_j(x_i)}{\partial x_1} + 2 \frac{\partial \phi_0(x_i)}{\partial x_2} \frac{\partial \phi_j(x_i)}{\partial x_2} - \phi_j(x_i) \right] dS \tag{61}$$

or in matrix form

$$\begin{aligned} \mathbf{K} &= (a_{jk}) \\ \mathbf{c} &= (\alpha_1, \ldots, \alpha_m)^T \\ \mathbf{b} &= (b_1, \ldots, b_m)^T \end{aligned}$$

$$\text{for} \qquad j, k = 1, \cdots, m$$

$$\mathbf{K} \cdot \mathbf{c} = \mathbf{b} \qquad . \tag{62}$$

From definition (60) it can be seen that the matrix $\mathbf{K}$ is symmetric, i.e. $a_{jk} = a_{kj}$.

3.6. 1D-Example for the Ritz Method with numbers

Let us consider the partial differential equation

$$\frac{\partial \sigma_{ij}}{\partial x_i} + \rho X_j = 0 \qquad . \tag{63}$$

In the 1D case expression (63) reduces according to figure (6) with no body forces to

$$E\frac{d^2u}{dx^2} = 0 \tag{64}$$

with the given Hooke law in one dimension

$$\sigma = E\epsilon \qquad and \qquad \epsilon = \frac{du}{dx} \tag{65}$$

$$\Rightarrow \quad \sigma \; dx = E \; du$$

where the E-Modul is assumed to be constant in time and space.

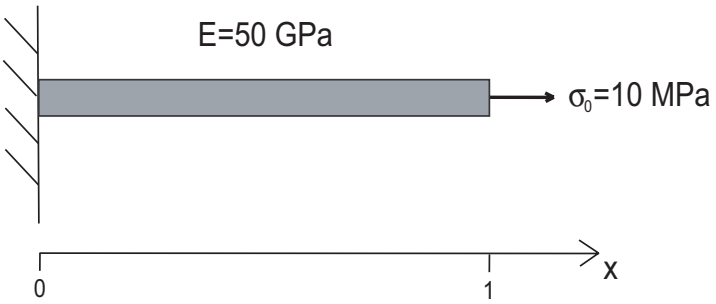


Figure 6: The bar is fixed to the wall on the left-hand-side. It has a constant Youngs Modul $E = 50$ GPa and is pulled by a force of $\sigma_0 = -10$ MPa on its right side. The length of the bar is one meter along the x axis.

According to figure (6) this is a bar of length one along the x-axis which is fixed on a wall on its left-hand-side. At the right side the bar is pulled by a force of $\sigma_0 = -10$ MPa. The Young Modulus of the bar is constant and has a value of $E = 50$ GPa. The boundary condition of the differential equation is $u(x = 0) = 0$ for the displacement at the wall. By inserting the given force boundary condition σ_0 at the position $u(x = 1)$ into the Hook's law we receive a second boundary condition in terms of the displacement

$$u(x = 1) = \frac{\sigma_0}{E} \tag{66}$$

According to expression (34) the boundary condition at $x = 1$ of the given 1D problem is

$$a_1 \frac{\partial u}{\partial x_1} n_1 = E\frac{\partial u}{\partial x} = \sigma_0 \tag{67}$$

The variational formulation of the differential equation (64) is

$$I(u) = \int_0^1 \frac{1}{2} E \left(\frac{du}{dx} \right)^2 dx - \sigma_0 u(x=1) \tag{68}$$

Let us now assume a linear approximation function $u_h(x)$ for $u(x)$

$$\begin{aligned} u_h(x) &= \alpha_0 + \alpha_1 x \\ \Rightarrow u_h(x) &= \alpha_1 x \end{aligned} \tag{69}$$

since $u_h(x=0) = \alpha_0 = 0$ due to the given boundary conditions. Inserting this approximation into the variational formulation gives

$$\begin{aligned} I(\alpha_1) &= \int_0^1 \frac{1}{2} E \left(\frac{d}{dx}(\alpha_1 x) \right)^2 dx - \sigma_0 \alpha_1 \\ &= \int_0^1 \frac{1}{2} E \alpha_1^2 dx - \sigma_0 \alpha_1 \end{aligned} \tag{70}$$

The first variation gives

$$\begin{aligned} \delta I(\alpha_1) &= \frac{dI(\alpha_1)}{d\alpha_1} = 0 \\ &= \int_0^1 E\alpha_1 \; dx - \sigma_0 \\ &= -E\alpha_1 - \sigma_0 \\ \\ \Rightarrow \alpha_1 &= \frac{\sigma_0}{-E} \end{aligned} \tag{71}$$

With the given values for E and the boundary conditions we then receive

$$\begin{aligned} \alpha_1 &= 0,0002 \\ \\ \Rightarrow u_h(x) &= 0,0002 \; x \end{aligned} \tag{72}$$

Since the stated linear 1D problem is solved exactly by a linear function, the chosen approximation $u_h(x)$ is the exact solution of the differential equation in expression (64).

3.7. Weighted residuals and Galerkin's Method

In case of a positive definite differential operator[6] a variational formulation always exists. But for other PDE's this is often not possible. Another approach has to be chosen, in order to get an appropriate for of an approximated formulation of the solution which can be used in a numerical method. This approach is the method of weighted residuals

[6] this is always true when the PDE is elliptical

of which the Galerkin[7] method is a special case. Again we choose an approximation $u_h(x_i)$ of the solution $u(x_i)$

$$\begin{aligned} u_h(x_i) &= \sum_{k=0}^{m} \alpha_k \ \phi_k(x_i) \\ &= \alpha_0\phi_0(x_i) + \sum_{k=1}^{m} \alpha_k \ \phi_k(x_i) \end{aligned} \tag{73}$$

where, again, $\phi_0(x_i)$ fulfils the boundary conditions. Substituting this approximation into expression (33) we obtain

$$\begin{aligned} R(x_i) &= \frac{\partial}{\partial x_1} a_1 \left(\sum_{k=0}^{m} \alpha_k \frac{\partial \phi_k(x_i)}{\partial x_1} \right) + \frac{\partial}{\partial x_2} a_2 \left(\sum_{k=0}^{m} \alpha_k \frac{\partial \phi_k(x_i)}{\partial x_2} \right) + \frac{\partial}{\partial x_3} a_3 \left(\sum_{k=0}^{m} \alpha_k \frac{\partial \phi_k(x_i)}{\partial x_3} \right) \\ &+ f \sum_{k=0}^{m} \alpha_k \ \phi_k(x_i) + g \end{aligned} \tag{74}$$

where $R(x_i)$ is the residual (defect) of the approximation. If $u_h(x_i)$ were the exact solution the residual $R(x_i)$ would equal zero. Hence, the nearer $R(x_i)$ tends toward zero, the better we can expect the approximation to be. The idea of the weighted residual method is that the residual is zero, but only in a weighted sense. The weight is provided by the $\phi_j(x_i)$. Every function $\phi_j(x_i)$ gives one equation. In this special case when the weighting function is the function used in the approximation the method is called Galerkin method[8].

The set of equations for the m unknown coefficients $\alpha_1, \ldots, \alpha_m$ follows from the requirement that the sum (=integral) of the product of $R(x_i)$ and $\phi_j(x_i)$ is identical to zero

$$\int_V R(x_i) \cdot \phi_j(x_i) \ dV = 0 \qquad for \qquad j = 1, \ldots, m \tag{75}$$

Mathematically spoken: If the inner product of two functions u and v is zero

$$< u, v > = \int_V u \cdot v \ dV = 0 \tag{76}$$

the functions u and v are orthogonal to each other. For our case this implies that the residual $R(x_i)$ should be orthogonal to all test functions $\phi_j(x_i)$.

$$< R(x_i), \phi_j(x_i) > = \int_V R(x_i) \cdot \phi_j(x_i) \ dV = 0 \tag{77}$$

That this requirement of orthogonality is good, is illustrated in figure 7. If the residual $R(x_i)$ is orthogonal to the function $\phi_j(x_i)$ it is the smallest.

[7]Boris G. Galerkin (1871-1945). Galerkin was a Russian engineer who published his first technical paper on the buckling of bars while imprisioned in 1906 by the Tzar in pre-revolutionary Russia, In many Russian texts the Galerkin method is known as the Bubnov-Galerkin method. Galerkin published a paper using this idea in 1915. The method was also attributed to I.G. Bubnov in 1913.

[8]If, for instance, the Dirac delta function is used as the weight function the method is called collocation

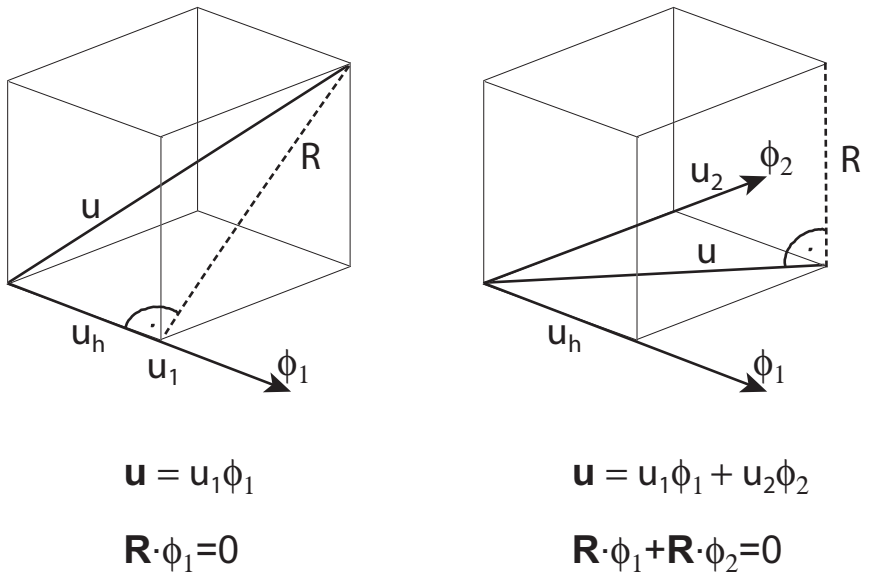


Figure 7: The sketch shows how the Galerkin method maintains orthogonality between the the residual vector R and the set of basis vectors ϕ_j. Two cases for one and two basis vectors are shown. In case of orthogonality the residual R is the smallest.

3.8. 2-D example for the Galerkin Method

Again as in the 2-D example for the Ritz Method, ($\beta = \gamma = 0$, $g = 1$, $f = 0$, $a_1 = 1$, $a_2 = 2$) the given PDE is

$$\frac{\partial^2 u}{\partial x_1^2} + 2\,\frac{\partial^2 u}{\partial x_2^2} + 1 = 0 \tag{78}$$

with the Dirichlet boundary condition $u = \chi$ on C_1 and the general Cauchy boundary condition on C_2

$$\frac{\partial u}{\partial x_1} n_1 + 2\,\frac{\partial u}{\partial x_2} n_2 = 0 \tag{79}$$

From expression (74) we receive with the above stated definitions for β, γ, g, f, a_1 and a_2 for this two dimensional problem

$$R(x_i) = 1 + \sum_{k=0}^{m} \alpha_k \frac{\partial^2 \phi_k(x_i)}{\partial x_1^2} + 2 \sum_{k=0}^{m} \alpha_k \frac{\partial^2 \phi_k(x_i)}{\partial x_2^2} \tag{80}$$

The equations for the coefficients $\alpha_1, \ldots, \alpha_m$ are received from the requirement that the integrals

$$\int_S R(x_i) \cdot \phi_k(x_i)\; dS \tag{81}$$

are identical to zero

$$\int_S \left[1 + \sum_{k=0}^{m} \alpha_k \left(\frac{\partial^2 \phi_k(x_i)}{\partial x_1^2} + 2\,\frac{\partial^2 \phi_k(x_i)}{\partial x_2^2} \right) \right] \cdot \phi_j(x_i)\; dS = 0 \tag{82}$$

Application of the partial integration, then applying the integral sentence of Gauss and exchanging of integration and summation as well as taking into consideration that $\alpha_0 = 1$ we receive

$$\begin{aligned} & \sum_{k=1}^{m} \alpha_k \int_S \left(\frac{\partial \phi_k(x_i)}{\partial x_1} \frac{\partial \phi_j(x_i)}{\partial x_1} + 2 \, \frac{\partial \phi_k(x_i)}{\partial x_2} \frac{\partial \phi_j(x_i)}{\partial x_2} \right) dS \\ + & \sum_{k=1}^{m} \alpha_k \int_S \left(\frac{\partial \phi_0(x_i)}{\partial x_1} \frac{\partial \phi_j(x_i)}{\partial x_1} + 2 \, \frac{\partial \phi_0(x_i)}{\partial x_2} \frac{\partial \phi_j(x_i)}{\partial x_2} - \phi_j(x_i) \right) dS = 0 \end{aligned} \tag{83}$$

This set of linear equations of j unknowns parameter α_j is identical to the set we received for that PDE with the Ritz Method in expression (59). Analog to the Ritz Method we can then define

$$a_{jk} := \int_S \left[\frac{\partial \phi_k(x_i)}{\partial x_1} \frac{\partial \phi_j(x_i)}{\partial x_1} + 2 \frac{\partial \phi_k(x_i)}{\partial x_2} \frac{\partial \phi_j(x_i)}{\partial x_2} \right] dS \tag{84}$$

$$b_j := - \int_S \left[\frac{\partial \phi_0(x_i)}{\partial x_1} \frac{\partial \phi_j(x_i)}{\partial x_1} + 2 \frac{\partial \phi_0(x_i)}{\partial x_2} \frac{\partial \phi_j(x_i)}{\partial x_2} - \phi_j(x_i) \right] dS \tag{85}$$

or in matrix form

$$\begin{aligned} \mathbf{K} &= (a_{jk}) \\ \mathbf{c} &= (\alpha_1, \ldots, \alpha_m)^T \\ \mathbf{b} &= (b_1, \ldots, b_m)^T \end{aligned}$$

$$\text{for} \qquad j, k = 1, \cdots, m$$

$$\mathbf{K} \cdot \mathbf{c} = \mathbf{b} \qquad . \tag{86}$$

From definition (84) it can be seen that the matrix $\mathbf{K}$ is symmetric, i.e. $a_{jk} = a_{kj}$.

4. Numerical solution - The Finite Element Method

Since we can not determine an approximation function by applying the Ritz or the Galerkin we now have to apply the Finite Element Method (FEM). This procedure includes the steps of discretization, finding an appropriate approximation function for this discretization, numerical integration and solving a linear set of equations (i.e. inverting a coefficient matrix).

4.1. First step into the world of Finite Elements

A global approximation function (polynomial, Fourier series etc.), as used in the Ritz Method, has three main disadvantages:

- The matrix $\mathbf{K} = (a_{jk})$ is full, i.e. it is difficult to invert.
- The approximation function is not very flexible, i.e. a local refinement in critical areas is almost impossible
- An approximation function is difficult to estimate where the geometry of the model area A and its boundary conditions are complicated. Especially a non-regular geometry of the model boundary is difficult to describe.

In order to overcome this "mission impossible" we subdivide the surface S with its border ∂S into subsurface (see figure 8). These subareas should not overlap each other and should have easy and smooth borders. In a two-dimensional case triangles and quadrangles are the most common definitions of subareas (finite elements). Their borders can be described by linear polygons. Instead of defining a complicated global approximation

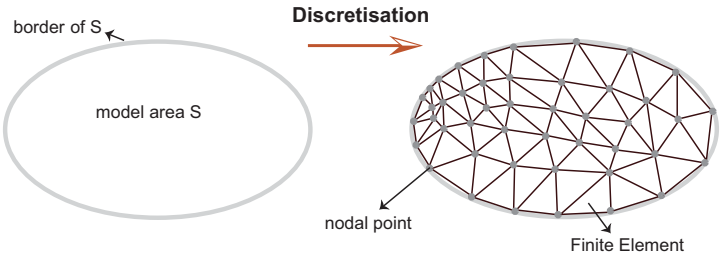


Figure 8: The model area S with its boundary ∂S is subdivided into subareas. These subareas can be triangles or quadrangles. These subareas are then called the Finite Elements. The interconnecting points between these Finite Elements at the corners are called nodal points or in short nodes.

function for the whole model surface S, we now only have to find an approximation function which describes the model behaviour in the subarea. We also reduce the continuity and differentiability requirements of this approximation. That means we only need to find a piecemeal continuous and piecemeal differentiable approximation function for

each subarea D_j with its border ∂D_j. Outside their subareas these local approximation functions are not defined. Mathematically spoken we have to fulfill the following two conditions:

$$(S \cup \partial S) \cong \bigcup_{j=1}^{m} (D_j \cup \partial D_j) \tag{87}$$

$$\phi_j(x_i) = \begin{cases} \text{approximation function} & \text{, if } \; x_i \in D_j \cup \partial D_j \\ 0 & \text{, else} \end{cases} \tag{88}$$

This is the basic idea which leads from the classical Ritz or Galerkin method to the Finite Element Method (FEM). The most convenient local approximation functions for these subareas are polynomials (linear, quadratic or cubic).

By using a linear approximation of the function $u(x_i)$ in a subarea we have to be "sure" that within the chosen subarea the variation of $u(x_i)$ can be approximated with a linear polynomial. If this is not the case we either have to choose a finer subdivision (later called mesh refinement) or a higher order of the polynomial approximation (cubic or quadratic) for the subarea. Of course the higher the number of subareas is and/or the higher the polynomial order is, the higher the computational costs are. But this will be discussed later. In a two-dimensional example the approximating function $u_h(x_i)$ for the subsurface S is

$$u_h(x_i) = \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 \tag{89}$$

$$u_h(x_i) = \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_1^2 + \alpha_4 x_2^2 + \alpha_5 x_1 x_2 + \alpha_6 x_1 x_2^2 + \alpha_7 x_2 x_1^2 \tag{90}$$

where the first function is a linear polynome and the second a quadratic polynome. The order of the approximation function is dependent upon the given problem and its discretization.

An important requirement is that continuity of the global solution $u_h(x_i)$ from one element to the neighboring elements must be fulfilled[9]. Elements which fulfill these continuity requirements are called conform. Besides this conformity requirement it is also required that the approximation functions do not change their geometrical forms under transformation (a straight line should remain a straight line after transformation) from one coordinate system to another. In this case the approximation functions are called (geometric) isotropic.

The following step is the basic idea of the Finite Element method. In order to fulfill the continuity requirements from one element to the next the usage of coefficients α_i is not suitable. It would be much more convenient to express the approximation function within the elements (I will use this term from now on in the text instead of the term

[9]Most of the continuity requirements are also obvious from a physical point of view. If, for instance, $u_h(x_i)$ describes the displacement field, a non-continuous behaviour from one element to the next would produce gaps.

subarea) through the function values of $u_h(x_i)$ at certain points of the elements[10]. These points are then called nodal points or nodes of the elements. In order to fulfill continuity from one element to the neighbouring one it is sensible to place these nodes at the corners of an element, i.e. a node is always shared by one or more elements. The values

$$u_k^{(e)}(x_i) \tag{91}$$

$$\begin{aligned} \text{where} \quad e &= 1, ..., n \quad \text{is the number of the element,} \\ p &= \text{the number of nodes per element and} \\ l, k &= 1, ..., p \quad \text{the node number of the element} \end{aligned}$$

of the approximation function at these nodes are called the nodal values of the node variable. With these nodal values the approximation function can be expressed as a linear combination of functions which use the nodal values of the nodal variables as their coefficients. For a two-dimensional element with p nodes we would then write:

$$u_h^{(e)}(x_i) = \sum_{k=1}^{p} u_k^{(e)} N_k^{(e)}(x_i) \tag{92}$$

Since this expression must be valid for all nodal values $u_k^{(e)}(x_i)$ the functions $N_k^{(e)}(x_i)$ must have the value one at the node k with the coordinates $x_{i,k}^{(e)}$ and zero at all others nodes of element e

$$N_k^{(e)}(x_{i,l}^{(e)}) = \begin{cases} 1 & \text{for } k = l \\ 0 & \text{for } k \neq l \end{cases} \tag{93}$$

If the nodal values are now globally (all elements) numbered from 1,...,m, the global approximation can be written as:

$$u_h(x_i) = \sum_{k=1}^{m} u_k N_k(x_i) \tag{94}$$

where $N_k(x_i)$ is the combination of all element shape functions $N_k^{(e)}(x_i)$ which have the value one at the nodal point k with the nodal value u_k.

The expression (94) which approximates the wanted function $u(x_i)$ has the same shape as the approximation function we used in the Ritz and Galerkin method. However, the values of the unknown coefficients α_j are now the values of the wanted function $u(x_i)$ at the node k, i.e. $u_k(x_i)$.

[10]Sometimes the partial derivatives of the approximation function $u_h(x_i)$ are taken. E.g. solving the displacement field, the partial derivative gives the strain field.

4.1.1. Transformation for a linear 1-D element

It is sensible to map the approximation function $u_h(x_i)$ onto a standard interval $[0, 1]$ in the ξ coordinate space. This mapping is done by defining the transformation formula. For a one dimensional linear approximation function the transformation formula from the x coordinate system into the ξ coordinate system is

$$x = x_j + (x_{j+1} - x_j)\xi \tag{95}$$

For the linear approximation

$$\phi(x) \;=\; \alpha_0 + \alpha_1 x \tag{96}$$

we receive in the ξ coordinates

$$\begin{aligned} \tilde{\phi}(\xi) &= \alpha_0 + \alpha_1[x_j + (x_{j+1} - x_j)\xi\ ] \\ &= \underbrace{\alpha_0 + \alpha_1 x_j}_{:=\tilde{\alpha}_0} + \underbrace{\alpha_1(x_{j+1} - x_j)}_{:=\tilde{\alpha}_1}\xi \end{aligned}$$

$$\Rightarrow \tilde{\phi}(\xi) \;=\; \tilde{\alpha}_0 + \tilde{\alpha}_1\xi \tag{97}$$

where x_j and x_{j+1} are the nodal values. Another possibility to represent the straight line is by defining so called shape functions. Since there is two nodes (or two unknown coefficients) for the linear approximation for the one dimensional problem, we need two shape functions N_1 and N_2

$$N_1(\xi) = 1 - \xi \qquad N_2(\xi) = \xi \tag{98}$$

in order to represent the straight line.

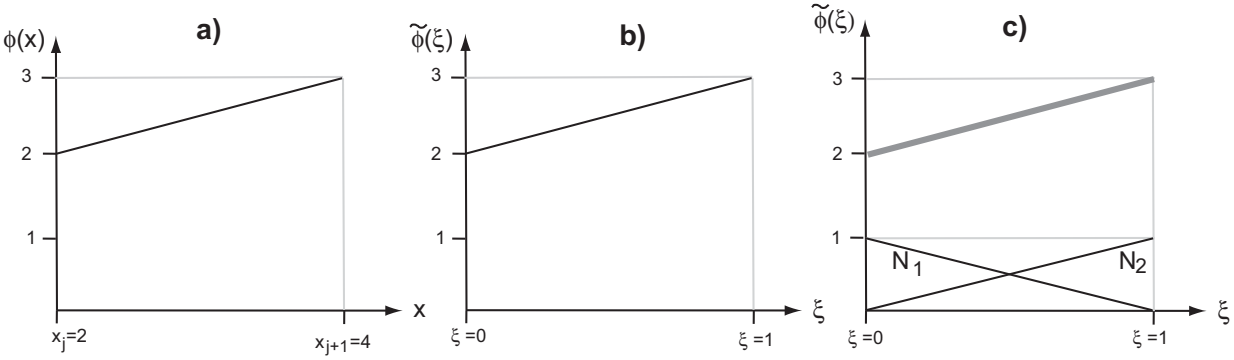


Figure 9: (a) straight line in the x coordinate space. (b) line after the transformation into the ξ coordinate space. (c) representation of this straight line by using the shape functions N_1 and N_2 in a linear combination (see expression 99)

By using these two shape functions the straight line can be represented as a linear combination of these two shape functions as shown in figure (9).

$$\begin{aligned} \tilde{\phi}(\xi) &= N_1\ \phi(x_j) + N_2\ \phi(x_{j+1}) \\ &= (1-\xi)\ \phi(x_j) + \xi\ \phi(x_{j+1}) \\ &= (1-\xi)\ u_j + \xi\ u_{j+1} \end{aligned} \tag{99}$$

where u_j and u_{j+1} are the values of the function u and the approximation u_h at the nodes j and $j+1$. Inserting the given values from figure (9) for $\phi(x_j)$ and $\phi(x_{j+1})$ we can write for expression (99)

$$\tilde{\phi}(\xi) = (1-\xi)\ 2 + \xi\ 3$$

$$\Rightarrow \qquad \begin{aligned} \tilde{\phi}(\xi=0) &= 2 \\ \tilde{\phi}(\xi=1) &= 3 \end{aligned}$$

The linear approximation in x coordinates for the given values is then

$$\phi(x) = 1 + 0,5x \tag{100}$$

and in ξ coordinates

$$\tilde{\phi}(\xi) = 2 + \xi \tag{101}$$

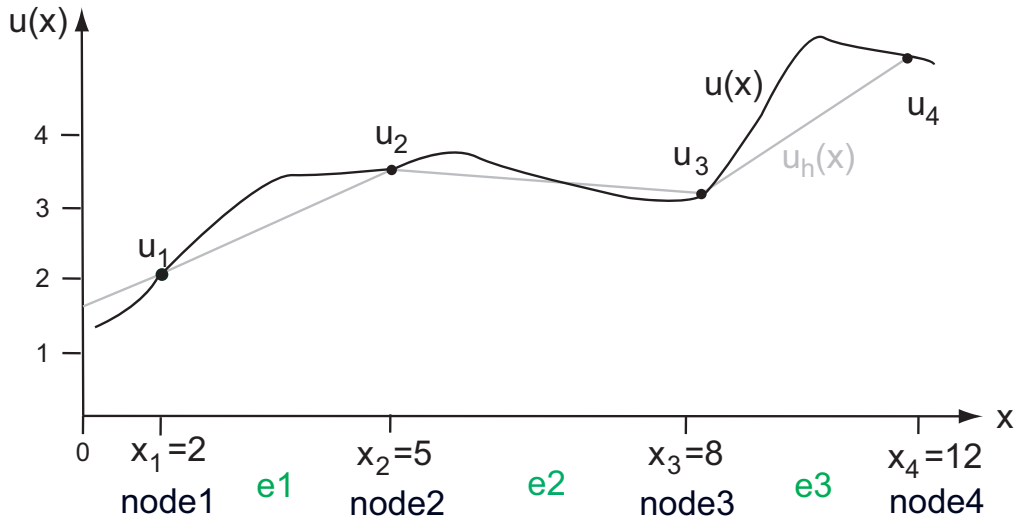


Figure 10: Subdivision of a bar into three elements with two nodes each. The exact solution $u(x)$ is approximated with linear segments. The values of the unknown coefficients α_i are the values of the function $u(x) = u_h(x)$ at the nodes. This preserves the continuity of the piecewise linear approximation $u_h(x)$.

4.1.2. Transformation of the derivatives for a linear 1-D element

For the example in the next chapter we also need the derivatives of the transformation formula (95) and some definitions:

$$\begin{aligned} x &= x_j + (x_{j+1} - x_j)\ \xi \\ \Rightarrow \frac{dx}{d\xi} &= x_{j+1} - x_j \\ &:= L_j \\ \Rightarrow dx &= L_j d\xi \end{aligned} \tag{102}$$

where L_j is the length of the subarea (the finite element). With the chain rule for derivatives we receive:

$$\begin{aligned} \tilde{\phi}'_j[\xi(x)] &= \frac{d\xi(x)}{dx}\frac{d\tilde{\phi}_j[\xi(x)]}{d\xi} \\ &= L_j^{-1}\frac{d\tilde{\phi}_j}{d\xi} \end{aligned} \tag{103}$$

4.2. 1-D Example

Let us now consider the following differential equation of second order

$$\frac{d}{dx}h(x)\frac{du}{dx} + f(x) = 0 \tag{104}$$

where $f(x)$ is the function for the body forces and the $h(x)$ describes the spatial variation of the material property (e.g. the Young Modulus E). The differential equation is defined within the interval

$$x \in [a, b]$$

$$u(a) = u_a \quad \text{and} \quad u(b) = u_b$$

According to the variation principal we receive the functional

$$I(u) = \int_a^b \left[\frac{1}{2}h(x)\left(\frac{du}{dx}\right)^2 - f(x)u\right]\, dx \qquad . \tag{105}$$

Let us now subdivide the area in subareas

$$a = x_1 < x_2 < ... < x_{m+1} = b$$

$$D_j := [x_j, x_{j+1}];\ j = 1, ..., m$$

and $u_h(x)$ the approximation of the wanted solution $u(x)$ Expression (105) can now be stated as the sum over the subinterval D_j

$$I(u_h) = \sum_{j=1}^{m} \int_{D_j} \left[\frac{1}{2} h(x) \left(\frac{du_h}{dx} \right)^2 - f(x) u_h \right] dx \tag{106}$$

and for each subinterval D_j we can then write

$$I_j(u_h) = \int_{x_j}^{x_{j+1}} \left[\frac{1}{2} h(x) \left(\frac{du_h}{dx} \right)^2 - f(x) u_h \right] dx \tag{107}$$

Mapping of the approximation function

$$u_h(x) = \phi(x) = \alpha_0 + \alpha_1 x \tag{108}$$

onto a standard interval $[0, 1]$ in the ξ coordinates space and inserting the approximation

$$u_h(\xi) = \tilde{\phi}(\xi) = \tilde{\alpha}_0 + \tilde{\alpha}_1 \xi \tag{109}$$

into expression (107) by using expressions (102) and (103) gives

$$\begin{aligned} I_j(\tilde{\alpha}_i) &= \int_0^1 \left[\frac{1}{2} h(\xi) \left(L_j^{-1} \frac{d\tilde{\phi}_j}{d\xi} \right)^2 - f(\xi) \tilde{\phi}_j \right] L_j d\xi \\ &= \frac{1}{2L_j} \int_0^1 h(\xi) \left(\frac{d\tilde{\phi}_j}{d\xi} \right)^2 d\xi - L_j \int_0^1 f(\xi) \tilde{\phi}_j d\xi \end{aligned} \tag{110}$$

Assuming that the intervals D_j are small enough the functions $f(x)$ and $h(x)$ is constant within the interval. Inserting the abbreviations

$$\bar{f} = \frac{1}{2}[f(x_j) + f(x_{j+1})] \tag{111}$$

$$\bar{h} = \frac{1}{2}[h(x_j) + h(x_{j+1})] \tag{112}$$

into expression (110) we receive

$$\begin{aligned} I_j(\tilde{\alpha}_i) &= \frac{\bar{h}}{2L_j} \int_0^1 \left(\frac{d\tilde{\phi}_j}{d\xi} \right)^2 d\xi - \bar{f} L_j \int_0^1 \tilde{\phi}_j d\xi \\ &= \frac{\bar{h}}{2L_j} \tilde{\alpha}_1^2 - \bar{f} L_j \frac{1}{2} (2\tilde{\alpha}_0 + \tilde{\alpha}_1) \end{aligned} \tag{113}$$

with

$$\begin{aligned} \int_0^1 \left(\frac{d\tilde{\phi}_j}{d\xi} \right)^2 d\xi &= \int_0^1 \tilde{\alpha}_1^2 d\xi \\ &= \tilde{\alpha}_1^2 \end{aligned} \tag{114}$$

and

$$\begin{aligned}\int_0^1 \tilde{\phi_j} d\xi &= \int_0^1 (\tilde{\alpha}_0 + \tilde{\alpha}_1 \xi) d\xi \\ &= \frac{1}{2}(2\tilde{\alpha}_0 + \tilde{\alpha}_1) \qquad .\end{aligned} \tag{115}$$

When u_j is the value of the continuous function u_h at the node j with its coordinates x_j with $j = 1, 2, ..., m$ the following relations are valid:

$$\begin{aligned}u_j &= u_h(x_j) \\ &= \phi_j(x_j) \\ &= \tilde{\phi_j}(\xi = 0) \\ &= \tilde{\alpha}_0\end{aligned} \tag{116}$$

$$\begin{aligned}u_{j+1} &= u_h(x_{j+1}) = \phi_j(x_{j+1}) \\ &= \tilde{\phi_j}(\xi = 1) \\ &= \tilde{\alpha}_0 + \tilde{\alpha}_1\end{aligned} \tag{117}$$

Stating expressions (116) and (117) in matrix gives

$$\begin{pmatrix} \tilde{\alpha}_0 \\ \tilde{\alpha}_1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} u_j \\ u_{j+1} \end{pmatrix} := \mathbf{p}\ \mathbf{u}^{(j)} \tag{118}$$

For $\tilde{\alpha}_1^2$ from expression (114) the matrix form is

$$\begin{aligned}\tilde{\alpha}_1^2 &= \begin{pmatrix} \tilde{\alpha}_0 \\ \tilde{\alpha}_1 \end{pmatrix}^T \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \tilde{\alpha}_0 \\ \tilde{\alpha}_1 \end{pmatrix} \\ &= \mathbf{u}^{(j)^T} \left[\mathbf{p}^T \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{p} \right] \mathbf{u}^{(j)} \\ &= \mathbf{u}^{(j)^T} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \mathbf{u}^{(j)} \\ &:= \mathbf{u}^{(j)^T} \mathbf{K}\ \mathbf{u}^{(j)}\end{aligned} \tag{119}$$

where $\mathbf{K}$ is the so called the stiffness matrix since here the physical properties are stored. For $0.5(2\tilde{\alpha}_0 + \tilde{\alpha}_1)$ from expression (115) the matrix form is

$$\begin{aligned}
\frac{1}{2}(2\tilde{\alpha}_0 + \tilde{\alpha}_1) &= \begin{pmatrix} \tilde{\alpha}_0 \\ \tilde{\alpha}_1 \end{pmatrix}^T \frac{1}{2} \begin{pmatrix} 2 \\ 1 \end{pmatrix} \\
&= \mathbf{u}^{(j)^T} \mathbf{p}^T \frac{1}{2} \begin{pmatrix} 2 \\ 1 \end{pmatrix} \\
&= \mathbf{u}^{(j)^T} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \frac{1}{2} \begin{pmatrix} 2 \\ 1 \end{pmatrix} \\
&= \mathbf{u}^{(j)^T} \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix} \\
&:= \mathbf{u}^{(j)^T} \mathbf{b}
\end{aligned} \tag{120}$$

where $\mathbf{b}$ is defined as

$$\mathbf{b} := \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix} \tag{121}$$

With the following two definitions

$$\omega_j := \frac{\bar{h}}{L_j} \tag{122}$$

$$\gamma_j := \bar{f} L_j \tag{123}$$

we can express equations (113) as

$$I_j = \mathbf{u}^{(j)^T} \frac{1}{2} \left(\omega_\mathbf{j} \, \mathbf{K}\mathbf{u}^{(j)} \right) - \mathbf{u}^{(j)^T} (\gamma_j \, \mathbf{b}) \tag{124}$$

Looking for a minimum of I_j the first variation leads to[11]

$$\delta I_j = \frac{\partial I_j}{\partial \mathbf{u}^{(j)}} = 0 \tag{125}$$

$$\Rightarrow \quad \omega_j \, (\mathbf{K}\mathbf{u}^{(j)}) - \gamma_j \, \mathbf{b} = 0 \tag{126}$$

As an example we assume that we have four elements. In matrix form we receive the following equations for each element[12].

$$e_1 : \qquad \omega_1 \begin{pmatrix} 1 & -1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{pmatrix} = \tfrac{1}{2} \begin{pmatrix} \gamma_1 \\ \gamma_1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \tag{127}$$

[11] The unknown variables are now the values of $\mathbf{u}^j$ at the nodes (nodal values). Therefore the variation has to be made towards $\mathbf{u}^j$.

[12] In order to be able to combine the single equations into a big set of equations the vectors $\mathbf{u}^{(j)}$ are filled up to a vector $\mathbf{u}$ with the missing $u_1 \cdots u_{j-1}, u_{j+2} \cdots u_{m+1}$and the matrixes $\mathbf{K}$ are filled with zeros in order to bring to the correct dimension of the problem which is a $(m+1 \times m+1)$-matrix.

$$e_2: \quad \omega_2 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 \\ \gamma_2 \\ \gamma_2 \\ 0 \\ 0 \end{pmatrix} \tag{128}$$

$$e_3: \quad \omega_3 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 \\ 0 \\ \gamma_3 \\ \gamma_3 \\ 0 \end{pmatrix} \tag{129}$$

$$e_4: \quad \omega_4 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 \\ 0 \\ 0 \\ \gamma_4 \\ \gamma_4 \end{pmatrix} \tag{130}$$

Combining these four expressions for the four elements through superposition we receive for the global stiffness matrix $\mathbf{K}^{(g)}$

$$\mathbf{K}^{(g)} = \begin{pmatrix} \omega_1 & -\omega_1 & & & 0 \\ -\omega_1 & \omega_1+\omega_2 & -\omega_2 & & \\ & -\omega_2 & \omega_2+\omega_3 & -\omega_3 & \\ & & -\omega_3 & \omega_3+\omega_4 & -\omega_4 \\ 0 & & & -\omega_4 & \omega_4 \end{pmatrix} \tag{131}$$

and

$$\mathbf{b}^{(g)} = \frac{1}{2}(\gamma_1, \gamma_1+\gamma_2, \gamma_2+\gamma_3, \gamma_3+\gamma_4, \gamma_4)^T \tag{132}$$

$$\mathbf{u}^{(g)} = (u_1, u_2, u_3, u_4, u_5)^T \tag{133}$$

This leads to the following set of equations.

$$\mathbf{K}^{(g)}\mathbf{u}^{(g)} = \mathbf{b}^{(g)} \tag{134}$$

The difference to the Ritz approach to the problem is obvious. The global stiffness matrix $\mathbf{K}^{(g)}$, which is the coefficient matrix, has now a so called bandwidth structure, i.e. only in the diagonal of the matrix and some parallel lines we have numbers. Most of the matrix consists of zeros which makes it very easy to invert the matrix and solve the problem. The term "stiffness matrix" expresses that in this matrix the physical behaviour of the model is stored.

5. Finite Element Method - Engineering approach

5.1. Engineering approach for the 1-D example

Another possibility to approach the Finite Element Method is to derive the equations received in the previous chapter by looking at an easy example. This is again a one-dimensional bar which is elongated along the x axis as shown in figure (11). This bar is divided into three segments of the length L_1, L_2, L_3 (Finite Elements) with three different Young's Modules E_1, E_2, E_3.

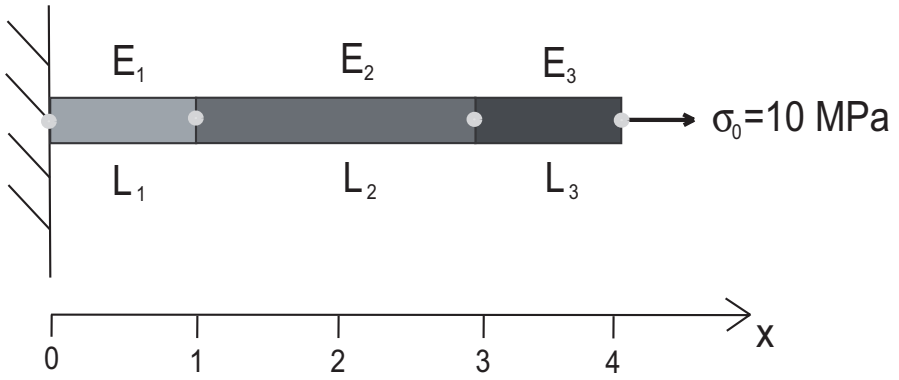


Figure 11: The bar is fixed on the left hand side at the wall. It is divided into three elements e_1, e_2, e_3 of the length L_1, L_2, L_3. Material properties within the elements are the Young's Modules E_1, E_2, E_3. The bar is pulled by a surface force of $\sigma_0 = 10MPa$ at its right side.

Figure (12) shows the Element e_2 with its displacements u_2 and surface force F_2 acting on its left side and the displacements u_3 and surface force F_3 acting on its right side.

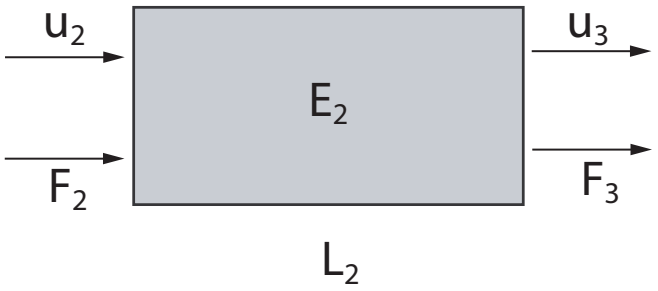


Figure 12: Sketch of the element e_2 with the length L_2 with the surface forces F_2, F_3 and displacements u_2, u_3 at its left and right hand side.E_2 is the Young's modulus.

The Hooke's law in each individual Finite Element can be written as

$$\sigma = E_i \epsilon \tag{135}$$

$$\epsilon = \frac{du}{dx} = \frac{u_i - u_{i+1}}{L_i}$$

where L_i is the length of each element. According to Hooke's law the surface force shown in figure (11) acting on the left-hand-side and the right-hand-side respectively can be written as

$$F_2 = \sigma_2 = \frac{E_2(u_2 - u_3)}{L_2} \tag{136}$$

$$F_3 = \sigma_3 = \frac{E_2(u_3 - u_2)}{L_2} \tag{137}$$

These two expressions can also be stated in matrix form and we receive

$$\frac{E_2}{L_2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} F_2 \\ F_3 \end{pmatrix} \Leftrightarrow \mathbf{K}^{(2)}\mathbf{u}^{(2)} = \mathbf{b}^{(2)} \tag{138}$$

$$\mathbf{K}^{(2)} = \begin{pmatrix} k_{11}^{(2)} & -k_{12}^{(2)} \\ -k_{21}^{(2)} & k_{22}^{(2)} \end{pmatrix} = \frac{E_2}{L_2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \tag{139}$$

For the elements e_1 and e_2 we can write in the same way

$$\frac{E_1}{L_1}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} F_1 \\ F_2 \end{pmatrix} \Leftrightarrow \mathbf{K}^{(1)}\mathbf{u}^{(1)} = \mathbf{b}^{(1)} \tag{140}$$

$$\mathbf{K}^{(1)} = \begin{pmatrix} k_{11}^{(1)} & -k_{12}^{(1)} \\ -k_{21}^{(1)} & k_{22}^{(1)} \end{pmatrix} = \frac{E_1}{L_1}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \tag{141}$$

$$\frac{E_3}{L_3}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} u_3 \\ u_4 \end{pmatrix} = \begin{pmatrix} F_3 \\ F_4 \end{pmatrix} \Leftrightarrow \mathbf{K}^{(3)}\mathbf{u}^{(3)} = \mathbf{b}^{(3)} \tag{142}$$

$$\mathbf{K}^{(3)} = \begin{pmatrix} k_{11}^{(3)} & -k_{12}^{(3)} \\ -k_{21}^{(3)} & k_{22}^{(3)} \end{pmatrix} = \frac{E_3}{L_3}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \tag{143}$$

where for each element $\mathbf{K}^{(e)}$ is the symmetric, quadratic element stiffness matrix, $\mathbf{u}^{(e)}$ the (displacement) vector of the nodal value (or in short node vector) and $\mathbf{b}^{(e)}$ the vector of the acting surface forces. By looking at node i which connects Element e_i and e_{i+1} we receive according to the equilibrium of the surface forces:

$$F_i - F_i^{(e)} - F_i^{(e+1)} = 0 \tag{144}$$

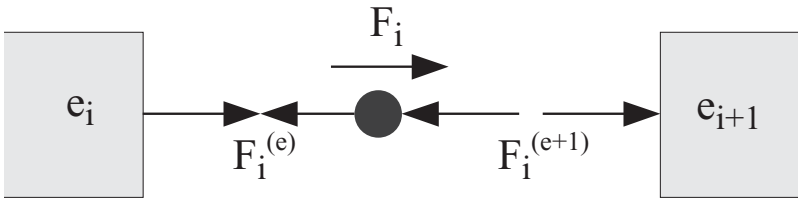


Figure 13: Sketch of the node i which connects the elements e_i and e_{i+1}. The surface force F_i at the node i has to be in equilibrium with the surface forces $F_i^{(e)}$ and $F_i^{(e+1)}$.

For the four nodes we then receive four equations

$$F_1^{(1)} = F_1 \quad \Leftrightarrow \quad k_{11}^{(1)} u_1 - k_{12}^{(1)} u_2 = F_1 \tag{145}$$

$$F_2^{(1)} + F_2^{(2)} = F_2 \quad \Leftrightarrow \quad -k_{21}^{(1)} u_1 + k_{22}^{(1)} u_2 + k_{11}^{(2)} u_2 - k_{12}^{(2)} u_3 = F_2 \tag{146}$$

$$F_3^{(2)} + F_3^{(3)} = F_3 \quad \Leftrightarrow \quad -k_{21}^{(2)} u_2 + k_{22}^{(2)} u_3 + k_{11}^{(3)} u_3 - k_{12}^{(3)} u_4 = F_3 \tag{147}$$

$$F_4^{(3)} = F_4 \quad \Leftrightarrow \quad -k_{21}^{(3)} u_3 - k_{22}^{(3)} u_4 2 = F_4 \tag{148}$$

These equations can be summarized in a global assembly and we receive the following global Finite Element equation in matrix form

$$\begin{pmatrix} k_{11}^{(1)} & -k_{12}^{(1)} & 0 & 0 \\ -k_{21}^{(1)} & -k_{22}^{(2)} + k_{11}^{(2)} & -k_{12}^{(2)} & 0 \\ 0 & -k_{21}^{(2)} & -k_{22}^{(2)} + k_{11}^{(3)} & -k_{12}^{(3)} \\ 0 & 0 & -k_{21}^{(3)} & k_{22}^{(3)} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} = \begin{pmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{pmatrix} \tag{149}$$

This can be written as

$$\mathbf{K}^{(g)} \, \mathbf{u}^{(g)} = \mathbf{b}^{(g)} \tag{150}$$

where $\mathbf{K}^{(g)}$ is the global stiffness matrix, $\mathbf{u}^{(g)}$ the global displacement vector and $\mathbf{b}^{(g)}$ the global load vector.

6. Special topics on the Finite Element Method

6.1. Do we need shape functions?

The term "shape function" is used for a different description of the approximation function for an element. But it is not compulsory to use shape functions explicitly in order to define a Finite Element. This is shown in the example in chapter 4.2 where no shape function was used for the development of the Finite Element equation.

However, in more complicated elements of higher order in 3-D it is much more convenient to integrate the equations when the approximation function within the element is described as a linear combination of the shape functions. The properties of the shape function are:

1. They can have the values one and zero.
2. They are only defined within the element. Outside the element their value is zero.

6.2. More definitions of Finite Elements

6.2.1. 1D Finite Element with a quadratic polynomial

By choosing a quadratic polynomial for the approximation function $u_h(x)$ we receive a third node, i.e. a third unknown coefficient, for the 1D Finite Element.

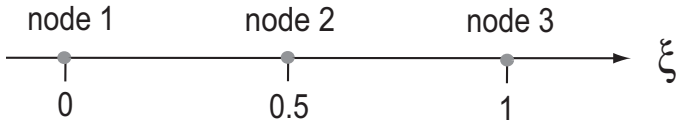


Figure 14: 1-D element with a quadratic polynomial, i.e. an element with three nodes.

$$\tilde{\phi}(\xi) = \alpha_0 + \alpha_1 \xi + \alpha_2 \xi^2 \tag{151}$$

From figure (14) we can express the values of the function $u_h(x)$ at the nodal points as

$$\begin{aligned} u_1 &= u_h(x_1) = \phi_j(x_1) = \tilde{\phi}_1(\xi = 0) \\ &= \alpha_0 \\ u_2 &= u_h(x_2) = \phi_j(x_2) = \tilde{\phi}_2(\xi = 0.5) \\ &= \alpha_0 + 0.5\alpha_1 + 0.25\alpha_2 \\ u_3 &= u_h(x_3) = \phi_j(x_3) = \tilde{\phi}_3(\xi = 1) \\ &= \alpha_0 + \alpha_1 + \alpha_2 \end{aligned} \tag{152}$$

Solving this for the α_i and writing the three expressions in matrix form we receive

$$\begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 4 & -1 \\ 2 & -4 & 2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} \tag{153}$$

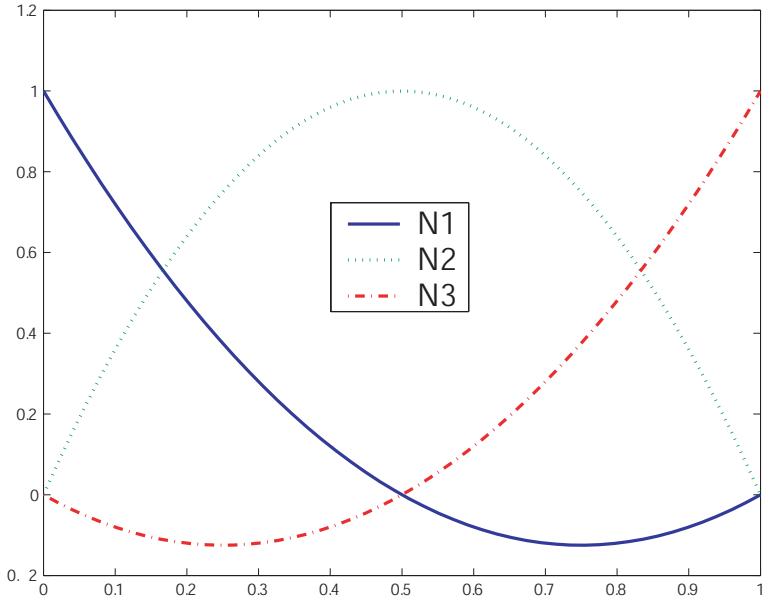


Figure 15: Graphs of the three shape functions for a 1-D element with a quadratic polynomial

The three shape functions are

$$N_1(\xi) = 1 - 3\xi + 2\xi^2 \tag{154}$$
$$N_2(\xi) = 4\xi - 4\xi^2 \tag{155}$$
$$N_3(\xi) = -\xi + 2\xi^2 \tag{156}$$

and $u_h(\xi)$ can the be expressed as a linear combination of the shape functions

$$u_h(\xi) = N_1\ u_1 + N_2\ u_2 + N_3\ u_3 \tag{157}$$

The three shape function N_1, N_2, N_3 are drawn in figure (15).

6.2.2. Linear transformation for Triangle Elements

Before we can have a look at the definitions for the triangle element, we have to state the mapping function for an arbitrary triangle T_i in the x_i coordinate system with the nodal points $P_1(x_1^{(1)}, x_2^{(1)}), P_2(x_1^{(2)}, x_2^{(2)})$ and $P_3(x_1^{(3)}, x_2^{(3)})$ onto a standard unit triangle T_0 in the ξ_i coordinate system (see figure 16).

The transformation formula for the mapping from the x_i coordinate system into the ξ_i coordinate system is

$$x_1 = x_1^{(1)} + [x_1^{(2)} - x_1^{(1)}]\xi_1 + (x_1^{(3)} - x_1^{(1)})\xi_2 \tag{158}$$

$$x_2 = x_2^{(1)} + [x_2^{(2)} - x_2^{(1)}]\xi_1 + (x_2^{(3)} - x_2^{(1)})\xi_2 \tag{159}$$

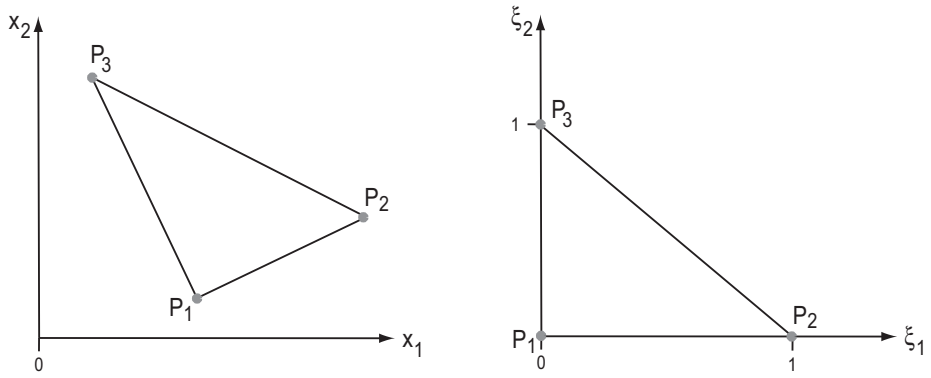


Figure 16: Mapping of an arbitrary triangle T_i in the x_i coordinate system with the nodal points $P_1(x_1^{(1)}, x_2^{(1)}), P_2(x_1^{(2)}, x_2^{(2)})$ and $P_3(x_1^{(3)}, x_2^{(3)})$ onto a standard unit triangle T_0 in the ξ_i coordinate system.

6.2.3. 2-D triangular elements with a linear polynomial

By choosing a linear polynomial for the approximation function $u_h(x)$ we receive for the 2D Triangular Element three nodes, i.e. three unknown coefficients α_i (see figure 16):

$$\tilde{\phi}(\xi) = \alpha_0 + \alpha_1\xi_1 + \alpha_2\xi_2 \tag{160}$$

Again we can express the values of the function $u_h(x)$ at the nodal points as

$$\begin{aligned} u_1 &= u_h(x_i^1) = \tilde{\phi}_1(\xi_1 = 0, \xi_2 = 0) \\ &= \alpha_0 \\ u_2 &= u_h(x_i^2) = \tilde{\phi}_2(\xi_1 = 1, \xi_2 = 0) \\ &= \alpha_0 + \alpha_1 \\ u_3 &= u_h(x_i^3) = \tilde{\phi}_3(\xi_1 = 0, \xi_2 = 1) \\ &= \alpha_0 + \alpha_2 \end{aligned} \tag{161}$$

Solving this for the α_i and writing the three expression in matrix form we receive

$$\begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} \tag{162}$$

The three shape functions are

$$N_1(\xi_1, \xi_2) = 1 - \xi_1 - \xi_2 \tag{163}$$

$$N_2(\xi_1, \xi_2) = \xi_1 \tag{164}$$

$$N_3(\xi_1, \xi_2) = \xi_2 \tag{165}$$

and $u_h(\xi)$ can be expressed as a linear combination of the shape functions

$$u_h(\xi_1, \xi_2) = N_1\ u_1 + N_2\ u_2 + N_3\ u_3 \tag{166}$$

Shape function N_1 is drawn as an example in figure (17).

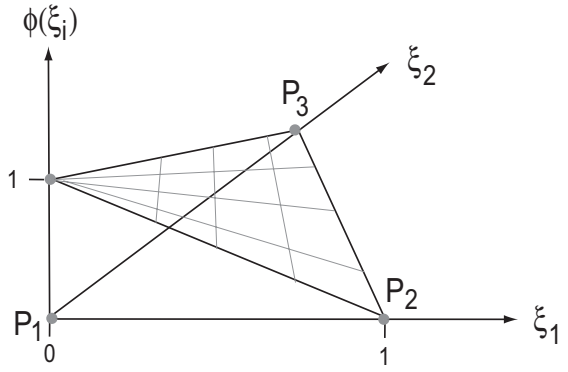


Figure 17: Shape function N_1 for a Triangular Element with a linear polynomial.

6.3. Classification of the various element types

According to space

1-D e.g. line, bar, cantilever
2-D e.g. triangle, quad
3-D e.g. Tetraeder, Hexaeder

According to the degree of the polynomial in 1-D

$$\begin{array}{ll} \alpha_0 + \alpha_1 x & \text{(linear)} \\ \alpha_0 + \alpha_1 x + \alpha_2 x^2 & \text{(quadratic)} \\ \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \alpha_3 x^3 & \text{(cubic)} \end{array} \tag{167}$$

According to completeness of the polynomial and properties

Isoparametric:	all elements where their transformation function preserves the geometrical shape
Lagrange:	complete polynomial
Serendipity:	incomplete polynomial
Conform:	every element which fulfills continuity of the node variables at the nodes

For example, if a quad element with a complete bi-quadratic polynomial is given it belongs to the Lagrange class. A quad element with a bi-quadratic polynomial belongs to the Serendipity class.

A. Mathematical rules and equations

A.1. Positive definite differential operator

Under certain conditions there is a relationship between a given differential equation and a variational formulation. Mathematically spoken: If a linear differential operator is positive definite, the differential equation can always be stated as a variational formulation. If for the inner product

$$< v, w > = \int_V v \cdot w \; dV \tag{168}$$

the following equations are true

$$\begin{aligned} < L \cdot v, w > = < v, L \cdot w > \quad &\Rightarrow \text{ L is self-adjoint, i.e. symmetric} \\ < L \cdot v, v > \quad > 0 \quad &\Rightarrow \text{ L is elliptic} \end{aligned} \tag{169}$$

the differential operator L is positive definite (e.g., the differential operator Laplace $\triangle$ and the differential operator Nabla ∇ operator are positive definite).

A.2. Calculus rules for the first variation

The calculus rules for the first variation are

$$\begin{aligned} \delta(u + v) &= \delta u + \delta v \\ \delta(u') &= (\delta u)' \\ \delta(u \cdot v) &= v \cdot \delta u + u \cdot \delta v \end{aligned} \tag{170}$$

Example from equation (35) in order to receive equation (36)

$$\begin{aligned} \delta\left[\left(\frac{\partial u}{\partial x}\right)^2\right] &= \delta\left[\frac{\partial u}{\partial x} \; \frac{\partial u}{\partial x}\right] \\ &= \frac{\partial u}{\partial x} \; \delta\frac{\partial u}{\partial x} + \delta\frac{\partial u}{\partial x} \; \frac{\partial u}{\partial x} \\ &= 2\left[\frac{\partial u}{\partial x} \; \delta\frac{\partial u}{\partial x}\right] \\ &= 2\left[\frac{\partial u}{\partial x} \; \frac{\partial}{\partial x}(\delta u)\right] \end{aligned}$$

A.3. Integration by parts

By integrating the product rule of differentiation one gets the integration by parts

$$\begin{aligned} \int_V (u \cdot v)' &= \int_V u' \cdot v + \int_V v' \cdot u \\ \Leftrightarrow \qquad \int_V u \cdot v' &= \int_V (u \cdot v)' - \int_V u' \cdot v \end{aligned} \tag{171}$$

The strike at the brackets in $(u \cdot v)'$ denotes the spatial derivation. Example in order to receive equation (37)

$$\int_V \left[a_1 \frac{\partial u}{\partial x_1} \frac{\partial}{\partial x_1}(\delta u) \right] \, dV \tag{172}$$

$$u = a_1 \frac{\partial u}{\partial x_1} \qquad \text{and} \qquad v' = \frac{\partial}{\partial x_1}(\delta u) \tag{173}$$

$$\Rightarrow \qquad u' = \frac{\partial}{\partial x_1}\left(a_1 \frac{\partial u}{\partial x_1} \right) \qquad\qquad v = \delta u$$

$$\Rightarrow \int_V \left[a_1 \frac{\partial u}{\partial x_1} \frac{\partial}{\partial x_1}(\delta u) \right] \, dV = \tag{174}$$

$$\int_V \left[\frac{\partial}{\partial x_1}\left(a_1 \frac{\partial u}{\partial x_1} \delta u \right) \right] \, dV - \int_V \left[\frac{\partial}{\partial x_1}\left(a_1 \frac{\partial u}{\partial x_1} \right) \delta u \right] \, dV \tag{175}$$

A.4. Gauss integral equation

The integral sentence of Gauss (divergence of a vector field) converts a volume integral to a closed surface integral.

$$\int_V \text{div } \mathbf{u} \, dV = \oint_S (\mathbf{u} \cdot \mathbf{n}) \, dS \tag{176}$$

A.5. Stokes integral equation

The integral sentence of Stokes (rotation of a vector field) converts a surface integral into a closed line integral integral.

$$\int_S (\text{rot}\mathbf{u} \cdot \mathbf{n}) \, dS = \oint_C \mathbf{u} \cdot \mathbf{dC} \tag{177}$$

For a plane case we receive

$$\text{rot}\mathbf{u} \cdot \mathbf{n} = \frac{\partial u_2}{\partial x_1} - \frac{\partial u_1}{\partial x_2} \qquad . \tag{178}$$

A.6. Stress definitions

The stress tensor can be transformed into a diagonal matrix. The remaining diagonal components are the principal stresses. σ_1, σ_2 and σ_3. For these it is defined that

$$\sigma_1 > \sigma_2 > \sigma_3 \qquad\qquad . \tag{179}$$

In Geoscience compressive stresses are positive. In engineering it is just the other way around!

The three invariants of the stress tensor expressed with the principal stresses are

$$\begin{aligned} I_1 &= \sigma_1 + \sigma_2 + \sigma_3 \\ I_2 &= -(\sigma_2\sigma_3 + \sigma_3\sigma_1 + \sigma_1\sigma_2) \\ I_3 &= \sigma_1\sigma_2\sigma_3 \end{aligned} \tag{180}$$

The mean stress (pressure) is

$$\sigma_m = p = \frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3) \tag{181}$$

The differential stress is defined as

$$\sigma_d = \sigma_1 - \sigma_3 \tag{182}$$

and the deviatoric stress as

$$\sigma'_{ij} = \sigma_{ij} - \sigma_m \tag{183}$$

In the presence of pore fluids, the pore fluid pressure σ_f, reduces the normal stresses. The effective normal stress is defined as

$$\sigma_{n_{eff}} = \sigma_n - \sigma_f \tag{184}$$

The brittle or plastic strength σ_y is a threshold value above which the rock material can not resist the stresses. It breaks (yields).

$$\sigma_{y_{max}} = \frac{1}{2}(\sigma_1 - \sigma_3) = C + \mu\sigma_n \tag{185}$$

where μ is the friction and C the cohesion. Both values are different depending if the yield strength is prescribed for existing faults (Amonton's law) or newly created fractures (Mohr-Coulomb criteria). For the friction on existing faults Byerlee (1966) determined C and μ in the laboratory with

$$\sigma_{y_{max}} = \frac{1}{2}(\sigma_1 - \sigma_3) = 0.85\sigma_n \tag{186}$$

when pressure is below 200 MPa (ca. 8 km depth) and

$$\sigma_{y_{max}} = \frac{1}{2}(\sigma_1 - \sigma_3) = 60MPa + 06\sigma_n \tag{187}$$

for deeper layers. In the literature this is stated as the Byerlee law.

A.7. Strain tensor

The strain tensor describes the deformation (Volume change and shape change) of a body.

$$\epsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) \tag{188}$$

After transformation into the principal system the three remaining diagonal components are the principal strains ϵ_1, ϵ_2 and ϵ_3 and it is defined

$$\epsilon_1 > \epsilon_2 > \epsilon_3 \tag{189}$$

The three invariants of the strain tensor expressed in the principal strains are

$$\begin{aligned} E_1 &= \epsilon_1 + \epsilon_2 + \epsilon_3 \\ E_2 &= -(\epsilon_2\epsilon_3 + \epsilon_3\epsilon_1 + \epsilon_1\epsilon_2) \\ E_3 &= \epsilon_1\epsilon_2\epsilon_3 \end{aligned} \tag{190}$$

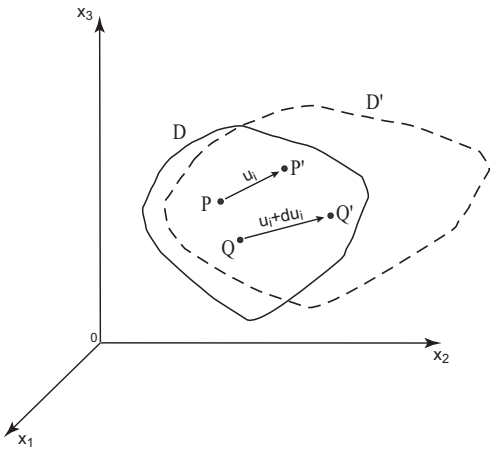


Figure 18: Displacement of two points P and Q of a body D towards the points P' and Q' of the new shape of the body D'.

In order to deduce the deformation tensor and extraction of the non deforming parts of translation and rotation two points P and Q according figure (18) of the body D are considered. After deformation towards P' and Q' the body has the new shape and position D'. The displacement of the two points P and Q is expressed through the displacement vectors u_i and $u_i + du_i$. In order to express $u_i(x)$ at an arbitrary location,

$u_i(x)$ in written in a Tayler expansion at point P.

$$\begin{aligned} u_i(Q) &= \sum_{n=0}^{\infty}(Q-P)^n \frac{u_i^n(P)}{n!} \\ &= u_i(P) + \left(\frac{\partial u_i}{\partial x_j}\right)_P \delta x_i \\ &= u_i(P) + \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j}\right)\delta x_i \\ &= u_i(P) + \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}\right)\delta x_i + \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\delta x_i \end{aligned} \tag{191}$$

The first term is the translation, the second the describes the antisymmetric rotation tensor and the third is the deformation tensor. The symmetric deformation tensor can be split into an isotropic part which is a pure volume change and shear part which reflects the shape change of the body.

B. List of useful books and papers

B.1. Books and paper quoted in the text

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C. English - German dictionary of mathematical expressions

acceleration	Beschleunigung
approximation	Näherung
assumption	Annahme
boundary condition	Randbedingung
boundary value problem	Randwertproblem
circular length	Bogenlänge
continuity	Stetigkeit
continuous function	stetige Funktion
defect	Residuum
deviatoric stress	deviatotrische Spannung
differentiability	Differenzierbarkeit
differential stress	differentielle Spannung
dilatation	Dehnung
ellipse	Ellipse
equilibrium of forces	Kräftegleichgewicht
equation of motion	Bewegungsgleichung
shape function	Form-Funktion
functional	Funktional
hyperbola	Hyperbel
initial value problem	Anfangswertproblem
integrand	Integrand
integration by parts	partielle Integration
mean stress	mittlere Spannung
one-differentiable	einfache Differenzierbarkeit
ordinary	gewöhnlich
parabola	Parabel
piecewise differentiable	stückweise differenzierbar
pressure	Druck
principal stress	Hauptspannung
quadrature	Quadratur
residual	Residuum
smoothness	Glattheit
space	Raum
spatial	räumlich
stationary	stationär, ortsfest
steady state	stationärer Zustand
steady flow	stationärer Fluss
strain	Deformation, Verformung
Taylor expansion	Taylor Entwicklung
unambiguousness	Eindeutigkeit
weighing function	Gewichtsfunktion
yield strength	Bruchfestigkeit, Scherfestigkeit